



Design, Development and Deployment of the Integrated Campaign Registry (ICR)

Technical Proposal

April 15th, 2026

Prepared for UNICEF

Annex B – Terms of Reference

LOT 1C – Web Applications and Open Source Related

Table of Contents

1. Understanding of UNICEF Requirements	5
1.1 Problem Statement	5
1.2 The Integrated Campaign Registry (ICR) Vision	5
1.3 Key Stakeholders and Ecosystem	6
2. Proposed Technical Approach	7
2.1 - ICR FHIR Implementation Guide Development	7
Deriving a Campaign Data Model in FHIR	7
Campaign Representation: CarePlan as the Architectural Foundation	8
Terminology Standards and Localization through ValueSets	10
Location Modeling, Administrative Hierarchies, and Geospatial Identity	10
2.2 IG Authoring, Validation, and Publication	11
Authoring in FSH with Standard Tooling	11
Technical Validation	11
Community Validation and Stakeholder Review	11
IG Publication and Release Lifecycle	12
2.3 ICR Reference Solution	13
Data Collection Layer	14
Data Transformation and Ingestion Layer	14
FHIR Store	14
Geospatial Visualization, Data Enrichment, and Microplanning	14
Integrate, Localize, Augment	15
Beyond Campaigns	15
3. Organization Expertise and Experience	16
3.1 Organization Profiles	16
Crosscut	16
3.2 WHO SMART Guidelines L3 Experience	17
3.3 HL7 FHIR Implementation Guide Experience	18
3.4 Digital Campaign Implementation Experience	18
Malaria IRS and ITN Campaigns	18
Immunization Campaigns	18
NTD Mass Drug Administration (MDA)	19
Understanding of Campaign Data Flows	20
3.7 Geospatial Microplanning Experience	20
3.8 Digital Public Good (DPG) Contributions	21
3.9 Production Data Integration Experience	22
3.10 Multi-Country Government Engagement	23
4. Implementation methodology/approach and workplan	25
4.1 Phase 1: Development of HL7 FHIR Implementation Guide (Months 1–2)	25

4.1.1 Approach	25
4.1.2 Deliverables	26
4.1.3 Key Dependencies	27
4.2 Phase 2: ICR Platform Development and Deployment (Months 3–6)	27
4.2.1 Approach	27
4.2.2 ICR Solution Package	28
4.2.3 Deliverables	29
4.2.4 Key Dependencies	29
4.3 Phase 3: Capacity Building and Training (Months 7–12)	29
4.3.1 Approach	29
4.3.2 Deliverables	30
4.3.3 Key Dependencies	30
Phases 4, 5, and 6: Concurrent Workstreams (Months 13–17)	30
4.4 Phase 4: Global and National Reporting Alignment (Months 13–17)	30
4.4.1 Approach	30
4.4.2 Deliverables	31
4.4.3 Key Dependencies	31
4.5 Phase 5: Systems Integration (Months 13–17)	31
4.5.1 Approach	31
4.5.2 Deliverables	32
4.5.3 Key Dependencies	32
4.6 Phase 6: Sustainability and Continuity (Months 13–17)	32
4.6.1 Approach	32
4.6.2 Deliverables	33
4.6.3 Key Dependencies	33
4.7 Limitations and Exclusions	34
7. Project Team	35
7.1 Organization Responsibilities	35
7.2 Consortium Structure	35
7.3 Key Personnel	36
Appendix 3	40
Proof of Compliance – FHIR Implementation Guide Criteria	40
Criterion 1 – Published, inspectable FHIR IG	40
Criterion 2 – Working FSH/SUSHI profiles (R4)	40
Required artifacts – source and rendered evidence	41
What each artifact demonstrates	41
Compilation evidence	41
Summary	42
Service Level Agreement (SLA)	43
1. Support Model	43

2. Response and Resolution Timelines	44
3. Issue Tracking, Reporting, and Resolution Management	44
4. Availability and Coverage	44

1. Understanding of UNICEF Requirements

1.1 Problem Statement

UNICEF is the integrating entity for campaign-based health delivery, supporting countries across immunization, polio, NTD mass drug administration, malaria prevention, and vitamin A supplementation. These programs frequently target the same communities and the same children, yet they have historically operated in isolation.

Campaign delivery has been verticalized for decades. Each campaign builds its own systems for planning, implementation, and reporting. Communities are remapped and people re-registered for every new round. Data from one program, whether household locations, population counts, or coverage tallies, is rarely available to inform the next, even when both campaigns serve the same villages.

This fragmentation is costly, particularly as resources tighten across global health. NTD programs are a clear example. NTD mass drug administration campaigns could leverage the household maps, population denominators, and geographic data that better-funded polio and immunization campaigns have already collected, but today's siloed systems make that impossible. Integration would benefit all sides: NTD programs gain data assets they cannot afford to build independently, while immunization programs gain a broader base of community-level intelligence that improves their own coverage tracking and microplanning.

Beyond inefficiency, fragmentation undermines data quality and timeliness. Campaign data is frequently paper-based, weakly linked to national HMIS, and invisible across programs. Governments and partners lack the cross-campaign visibility needed to identify coverage gaps, coordinate co-delivery, and hold programs accountable.

The opportunity is to shift campaigns from isolated, disposable data events to contributors to a cumulative, reusable corpus of public health intelligence, transforming each campaign's data collection costs from a one-off expenditure into an investment that compounds across programs and over time.

1.2 The Integrated Campaign Registry (ICR) Vision

UNICEF seeks to realize this shift through the Integrated Campaign Registry (ICR): a standards framework and reference implementation that enables public health campaigns to share, exchange, and reuse data and metadata in a structured, interoperable way. The ICR represents a foundational shift in how campaign data is managed, establishing an open standard for how campaign information is stored and exchanged so that what one campaign collects can be repurposed by the next rather than re-collected from scratch.

At its core, the ICR acts as a metadata and data backbone for campaign integration. It receives data from whichever collection tools countries already use (ODK, DHIS2 Tracker, CommCare, and others), normalizes and harmonizes that data, and makes it available for planning, monitoring, reporting, and reuse across programs. The ICR is designed to align with national HMIS and WHO reporting requirements, strengthening rather than duplicating existing health system infrastructure.

The ICR will be built on HL7 FHIR (Fast Healthcare Interoperability Resources) standards to ensure global interoperability and long-term sustainability. As outlined in the TOR, its architecture spans seven interconnected components: a FHIR Implementation Guide defining the data model and interoperability rules; a core registry database; integration and data flow tools bridging diverse source systems to the registry; a standardized data model covering patients, households, locations, services delivered, and campaign metadata; an analytics layer enabling data to flow into warehouses for monitoring and decision-making; a data enrichment layer integrating with tools like the WHO Geospatial Microplanner; and a capacity building framework including training, documentation, and a replication toolkit.

More specifically, we understand the objectives of this assignment to be:

- Design and develop a scalable, interoperable ICR built on HL7 FHIR standards and aligned with national and global reporting requirements.
- Facilitate the integration and repurposing of data and metadata from existing campaign tools across immunization, NTD mass drug administration, and vitamin A supplementation, avoiding duplication and enabling reuse across programs.
- Strengthen linkages between community-level reporting and national HMIS, ensuring timely and accurate data flow and improved reporting of UNICEF-supported MDA campaigns to relevant country and global platforms.
- Enhance the capacity of governments and partners in two pilot countries to use the ICR for planning, monitoring, and reporting, supporting long-term institutionalization and sustainability.

Our proposal responds to the full scope of this vision. In the sections that follow, we describe our technical approach to each component, grounded in our direct experience building FHIR-based health information systems at national scale and supporting campaign digitalization across multiple countries and program areas.

1.3 Key Stakeholders and Ecosystem

This initiative sits within an ecosystem of stakeholders and digital platforms that the ICR must integrate with and support.

- UNICEF is the system owner, with HQ teams across NTD, Community Health, Digital Health, and DAPM, along with Regional and Country Offices in the pilot countries leading coordination with governments and partners.
- Ministries of Health in each pilot country are the primary government counterparts responsible for NTD programs, immunization, and national HMIS. The ICR must embed within their reporting cycles and digital health architectures.
- WHO, including ESPEN, immunization and polio programs, and the Geospatial Microplanner team. The ICR must align with JAP reporting formats and support data exchange with WHO tools.
- Gavi, whose reporting requirements for immunization campaigns must be supported.
- Implementing partners, including community health organizations and NGOs engaged in NTD MDA, immunization, and nutrition campaigns.
- Digital platforms already embedded in country workflows, including DHIS2, ODK, OpenSRP, ESPEN tools, and the WHO Geospatial Microplanner. The ICR must be interoperable with all of these.

2. Proposed Technical Approach

2.1 - ICR FHIR Implementation Guide Development

A FHIR Implementation Guide (IG) is a specification that defines how FHIR resources should be structured and constrained for a specific use case. It serves as a shared rulebook, ensuring that different systems built by different teams can exchange the same data reliably.

Deriving a Campaign Data Model in FHIR

Public health campaigns present a fundamental modeling challenge in FHIR. The FHIR specification was designed around clinical care encounters: a patient presents at a facility, receives a service, and a record is created. Campaigns invert this model. They are population-level, geographically driven, time-bounded events in which teams move outward into communities to deliver services at scale. There is no native Campaign resource in FHIR or CampaignEvent. This means the ICR IG must identify which existing FHIR resources can be profiled and extended to carry campaign semantics faithfully, and it must do so in a way that is architecturally sound and endorsed by the broader FHIR community.

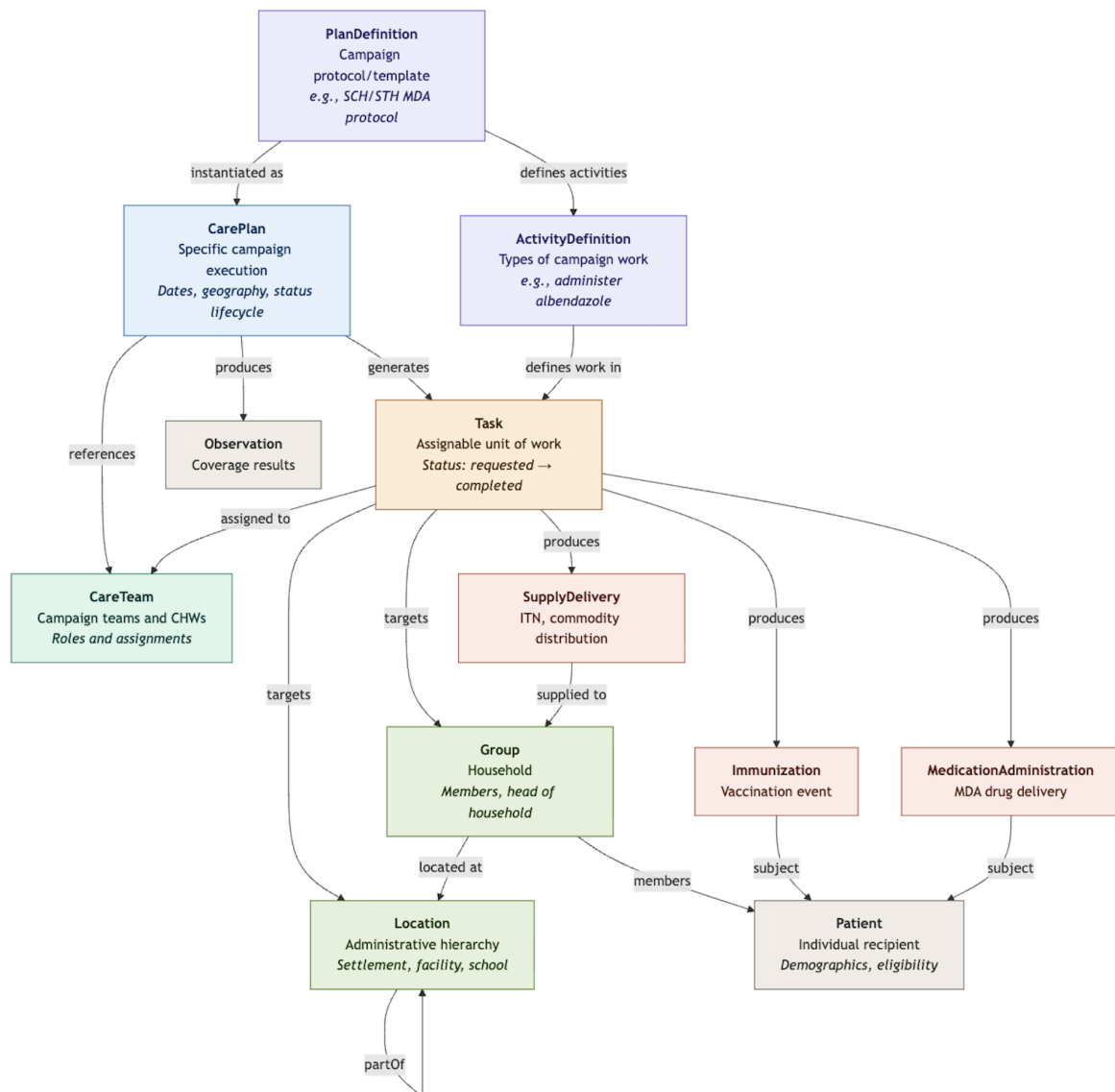
The ICR FHIR IG will be grounded in operational reality. We will begin by reviewing data from 3–4 existing public health campaigns in collaboration with UNICEF, including NTD mass drug administration, immunization, polio, etc. to extract the implicit data model that campaigns already use in the field. For each campaign dataset, we will catalogue the data elements captured, identify common and divergent structures across campaign types, and synthesize a canonical data model covering the core entities: target populations and households, administrative and service delivery locations, services delivered, commodities distributed and consumed, campaign teams and their assignments, and campaign metadata including dates, target populations, implementing partners, and coverage results.

We have navigated this exact challenge before. When we needed to represent households for community health workflows, a concept with no native FHIR support, we identified the closest base resources (the Group resource, linked to Location for the physical dwelling), defined the necessary extensions, and worked through the HL7 community process to validate the approach. That household representation has since been adopted and extended within the FHIR ecosystem. We will follow the same methodology for campaign data, engaging the FHIR community early and iteratively to ensure the ICR IG has both technical legitimacy and a path toward long-term adoption.

As part of this data modeling work, we will define which components need to be captured and reported in real-time during an active campaign versus those that are reconciled at campaign close. Real-time components would typically include doses or commodities administered, locations visited, and progress against targets, enabling campaign managers to monitor coverage and redeploy teams while the campaign is still underway. Post-campaign components would include reconciled stock counts, final coverage calculations, data quality reviews, and consolidated reporting aligned to WHO Joint Application Package (JAP) forms for example. The data model will be designed so that a single structure supports both operational monitoring during the campaign and formal reporting afterward, rather than requiring separate systems for each.

Campaign Representation: CarePlan as the Architectural Foundation

Based on our analysis of campaign workflows and FHIR's resource model, we believe the CarePlan resource provides the most natural and powerful foundation for representing public health campaigns. In clinical FHIR, a CarePlan represents a coordinated set of activities to address a health concern for a patient or group; a campaign is the same concept at population scale. The campaign data model we will develop and validate builds on this foundation using several interconnected FHIR resources. PlanDefinition would serve as the reusable campaign protocol, the template that defines what a given type of campaign involves (drugs, target age groups, activity sequence) and that can be instantiated consistently across geographies and time periods. Each specific campaign execution would be represented as a CarePlan referencing its parent PlanDefinition and carrying the campaign-specific details: dates, target geography, target population, and a status lifecycle from draft through active to completed. Within the CarePlan, ActivityDefinition resources define the discrete types of work (e.g., "administer albendazole to children aged 5–14," "distribute ITNs to households"), while Task resources make the plan operational. Each Task is an assignable, trackable unit of work tied to a specific Location (a settlement, a school, a set of households), assigned to a member of the CareTeam, and carrying a status that tracks progress from requested through completed. Individual service delivery events would be recorded using Immunization (for vaccinations), MedicationAdministration (for MDA drugs), and SupplyDelivery (for commodity distribution), each linked back to the relevant Task and Location. Households would be represented using the Group resource, with members referencing individual Patient records where person-level data is collected, linked to the Location where the household is situated.



HL7 FHIR CarePlan for modeling Campaigns

This architecture has a critical additional benefit: it unifies campaign planning and campaign execution within a single data model. A CarePlan that begins as a microplan, defining what needs to happen, where, by whom, and with what supplies, evolves into a campaign execution record as Tasks are completed and coverage data accumulates. This directly supports the project's objective of linking campaign execution ultimately to microplanning, and it provides a clean integration path with tools like the WHO Geospatial Microplanner, where CarePlans, Tasks, and Locations can flow bidirectionally as FHIR resources. Design decisions around granularity (whether Tasks are assigned at the village level or down to individual households), performance at national scale, and real-time versus post-campaign reconciliation are precisely the types of questions we will work through with the FHIR community and validate against real campaign data during the IG development process.

Terminology Standards and Localization through ValueSets

Since the ICR will span multiple campaign types, including immunization, NTD MDA, malaria prevention, and vitamin A supplementation, each with its own product formularies, target population definitions, and reporting categories. The IG must define terminology bindings that are internationally standardized yet locally adaptable. For immunization events, CVX (CDC's Vaccines Administered code system) is well-established and already supported by FHIR's Immunization resource. For pharmaceutical products used in MDA campaigns (praziquantel, albendazole, ivermectin, mebendazole), the WHO ATC (Anatomical Therapeutic Chemical) classification provides an internationally recognized coding system appropriate for a global IG. For commodity identification in supply chain tracking, including specific drug formulations, ITN products, and IRS chemicals, GS1 GTINs offer product-level identification, while broader supply chain workflows can reference WHO Essential Medicines List categories.

The IG will define standardized code lists (ValueSets) for key data elements. For core campaign types (vaccination, MDA, ITN distribution, IRS, vitamin A supplementation), implementations will be required to use the codes defined in the IG to ensure consistency. For drug and vaccine products, implementations will be expected to use international codes (such as CVX or ATC) when applicable, but will be able to add local codes for country-specific products. The IG will include mapping resources (ConceptMaps) that show how local codes relate back to the international standards. This approach allows countries to incorporate their own product names and registration numbers while keeping data comparable across countries and compatible with the international IG.

Location Modeling, Administrative Hierarchies, and Geospatial Identity

The FHIR Location resource will require the most customization of any resource in the ICR IG. Campaign data is fundamentally organized by geography: administrative hierarchies (country → region → district → sub-district → ward → village/settlement), service delivery points (health facilities, schools, community distribution sites), and individual structures (households). Every aspect of campaign planning, execution, and reporting depends on a well-defined and performant location model.

FHIR Location supports parent-child relationships, allowing locations to be nested within administrative hierarchies (e.g., a village within a ward within a district). However, the deep nesting typical of campaign countries, often six or more levels, creates performance challenges that must be carefully designed around to keep mobile and web applications responsive. This will be a particular area of focus during IG development, drawing on our experience with nested location hierarchies in production FHIR implementations at national scale.

The Location resource will also need to be extended to support not just point coordinates (GPS lat/long) but geographic boundary data such as district polygons, settlement areas, and facility catchment zones. We will define a GeoJSON extension on the Location resource to carry these shapes. Stable identification of locations across campaigns is equally important. The IG will support multiple identifier systems: P-codes (the OCHA standard for administrative boundaries), Overture Maps GERS IDs (a global, open identifier system covering over 2.6 billion buildings and 64 million places, offering a way to consistently identify the same household or settlement across

campaigns), and country-specific identifiers such as national facility registry codes. This multi-identifier approach ensures locations can be reliably linked regardless of which system a given country uses.

2.2 IG Authoring, Validation, and Publication

Authoring in FSH with Standard Tooling

The ICR IG will be authored in FHIR Shorthand (FSH), the HL7-endorsed domain-specific language for IG development. FSH is concise, human-readable, and, critically, text-based, which means all IG source files are managed under version control in a public GitHub repository with full change history, branching, and collaborative review through pull requests. The FSH source files are compiled by SUSHI into FHIR R4 conformance artifacts (StructureDefinitions, ValueSets, CodeSystems, Extensions, ConceptMaps, and example instances), which are then rendered into a browsable HTML publication by the HL7 IG Publisher. This toolchain (FSH → SUSHI → IG Publisher) is the standard used by the vast majority of contemporary FHIR IGs, including WHO SMART Guideline IGs, and ensures that the ICR IG is structurally consistent with the broader FHIR ecosystem.

Technical Validation

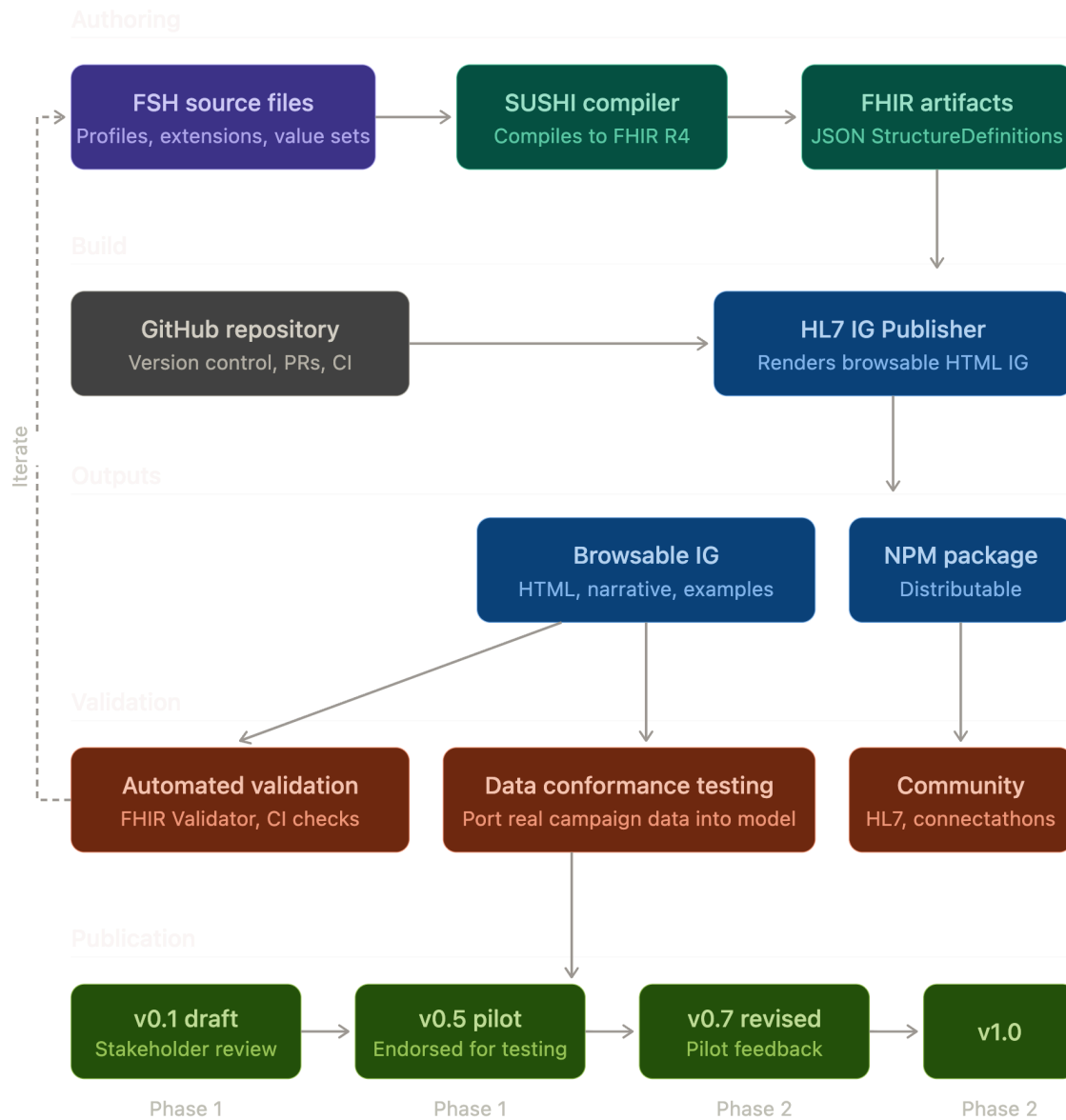
Validation will happen at multiple levels. The FHIR toolchain (SUSHI and the IG Publisher) automatically checks that all profiles, ValueSets, and examples conform to the FHIR R4 specification and are internally consistent. We will configure continuous integration so that every change to the IG is automatically built and validated, surfacing errors immediately. Beyond this automated validation, we will perform data conformance testing by taking real campaign datasets and attempting to convert them into the ICR FHIR model, identifying where the model fits cleanly and where it needs adjustment. This ensures the IG works with real-world data, not just in theory.

Community Validation and Stakeholder Review

Beyond technical validation, the IG must be reviewed by both the FHIR standards community and the public health stakeholders who will use it. We will engage the HL7 community through established channels, including chat.fhir.org, relevant working group calls, and Connectathon events where implementers test IGs against real systems. This builds consensus for the campaign-related extensions to be adopted broadly, as happened with the household representation we developed for community health.

On the stakeholder side, each draft will be circulated to MoH, UNICEF, WHO, and implementing partners as a versioned, browsable IG so reviewers can inspect actual profiles and examples rather than static documents. Feedback will be tracked in GitHub and incorporated into subsequent revisions.

IG Publication and Release Lifecycle



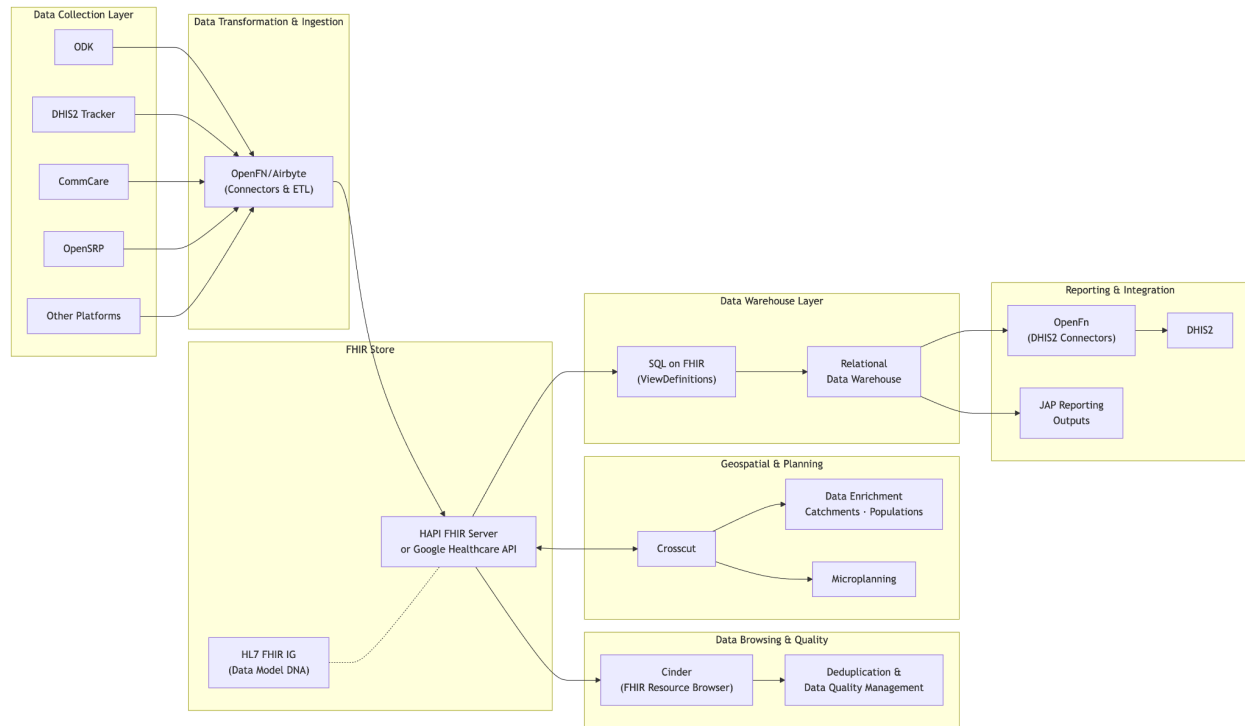
IG Publication Lifecycle

The IG will follow FHIR community publication conventions, with each formal release published as a versioned, browsable IG with a stable URL, accompanied by an NPM package that enables other IGs and FHIR tools to declare a dependency on it. All source, generated artifacts, and published output will be hosted on GitHub under an open-source license, ensuring transparency and community contribution.

We envision the following release cadence aligned to the project phases: a v0.1 draft IG produced during Phase 1 and circulated for stakeholder and community review; a revised draft incorporating that feedback and endorsed for use in the initial pilot; updated versions

incorporating learnings from pilot country deployments during Phase 2; and a v1.0 release representing the validated, production-ready IG. Each release will include a changelog and maintain backward compatibility where possible, with clear migration guidance where breaking changes are necessary.

2.3 ICR Reference Solution



Proposed ICR Package Architecture

The ICR will be conceived not as a software platform but as a **packaged reference solution**, an open-source, standards-based blueprint that demonstrates how campaign data can be integrated, harmonized, and reused across programs. The reference solution will be composed of interchangeable open-source components that can be substituted or adapted based on country context, existing infrastructure, and institutional preferences. What will remain constant across any implementation is the HL7 FHIR Implementation Guide, which serves as the DNA of the entire system. The IG will define the data model, the structure and semantics of campaign resources, and any conformant implementation, regardless of the specific software stack, will need to adhere to it.

Our proposed reference ICR solution will be organized into the following functional layers:

Data Collection Layer

The ICR will not prescribe a single data collection tool. Campaigns will continue to use the platforms already embedded in their workflows, including ODK, DHIS2 Tracker, CommCare, OpenSRP, and others. The ICR will be designed to receive data from any of these systems, meeting

countries where they are rather than requiring adoption of a new collection tool. This is a foundational design principle: enhance existing country systems, not replace them.

Data Transformation and Ingestion Layer

To bridge the gap between the diverse data formats produced by collection platforms and the normalized FHIR data model required by the ICR, we will implement a middleware layer using OpenFn or Airbyte as the primary integration engine. Using a tool like OpenFn, we will build connectors that extract data from source systems, transform it into FHIR-conformant resources as defined by the IG, and load it into the FHIR store. The connectors developed through this project will be openly available and documented as part of the replication toolkit, allowing other implementers to adapt them for additional source systems.

FHIR Store

The core registry database will be a FHIR R4-compliant server, with the open-source HAPI FHIR as the primary reference implementation and the Google Healthcare API as an alternative for cloud-hosted deployments. This choice is well aligned with the direction many countries across Africa are already taking: HAPI is increasingly being adopted as the default FHIR server and shared health record store within national health information exchange architectures, making the ICR immediately compatible with the infrastructure these countries are building. The FHIR store will serve as the canonical repository. Once data passes through the transformation layer and is loaded, it will exist in a standardized, interoperable format regardless of its origin. The store will expose RESTful APIs that enable all downstream components, including browsing, analytics, planning, and reporting, to consume data consistently.

Data Browsing and Quality Management. We will integrate Cinder (<https://cinder.recode labs.org>), a FHIR resource browser developed by Recode Labs as part of the OHS project, to provide a user-facing interface for exploring and managing the data held in the FHIR store. Cinder will enable users to browse resources, inspect records, and navigate the relationships between campaign entities such as patients, locations, encounters, and observations. Critically, Cinder will also support data quality management functions, including deduplication, validation, and anomaly detection, which will be a major workstream in this project. Given that the ICR will receive data from multiple campaigns using different collection tools, deduplication of patients, households, and locations across sources will be essential to the integrity of the registry. Cinder will provide the tooling needed to identify, review, and resolve duplicate records and other data quality issues.

Geospatial Visualization, Data Enrichment, and Microplanning

The ICR will integrate with Crosscut, which will serve as the geospatial and planning layer of the solution. Under this project, Crosscut will be adapted to consume FHIR data from the ICR to visualize health facilities, service delivery points, settlements, and other campaign-relevant locations on a map. Beyond visualization, Crosscut will support data enrichment workflows. For example, if the ICR contains a set of health facilities or villages identified as campaign targets, those entities can be imported into Crosscut to define catchment areas, assign population estimates, calculate distances, and perform other spatial analyses needed for microplanning. Once enriched, this data will be pushed back into the ICR, creating a bidirectional flow that strengthens both the registry and the planning process. This integration directly supports the TOR

requirement for two-way integration with microplanning tools and will ensure that ICR data feeds meaningfully into campaign planning, not just reporting.

FHIR-to-Data-Warehouse Layer. To make the structured FHIR data accessible for broader analytical use, we will implement a SQL on FHIR layer that will transform FHIR resources into flat, queryable tables in a relational data warehouse. This transformation will be driven by FHIR ViewDefinition resources, themselves part of the IG, which will formally define how FHIR resources should be projected into tabular views. By including ViewDefinitions in the IG, we will ensure that the analytical layer is as standardized and portable as the data model itself: any implementer using the ICR IG will be able to generate the same warehouse schema from their FHIR store. We have already validated this methodology in prior work in Uganda with the MoH and UNICEF and will implement it as a core component of the reference package.

DHIS2 Integration and Reporting. From the data warehouse, we will use OpenFn to build connectors that synchronize ICR data with DHIS2, supporting both import and export of aggregate and tracker data. This approach, from FHIR store to data warehouse to DHIS2, has been proven in our prior implementations and offers significant advantages over direct FHIR-to-DHIS2 mapping. The warehouse layer will provide a stable, well-understood relational structure that simplifies the alignment of ICR data elements with DHIS2 data sets and program indicators. This same pathway will support the generation of JAP-aligned reporting outputs.

Integrate, Localize, Augment

The ICR is designed to work with, not replace, the tools countries are already using. Where national systems have existing platforms, APIs, and workflows, the ICR will plug into them through interoperability layers built on HL7 FHIR. Where gaps exist, it will fill them in alignment with national digital health architectures. Because no two country contexts are the same, the registry is built to be localized—adapting ValueSets, workflows, and data models to each country's health system rather than imposing a one-size-fits-all approach.

Beyond Campaigns

A core ambition of this work is to establish a common data model backbone that starts with campaigns but extends to broader health system use. A location registry built for NTD mass drug administration, for example, should be equally useful for routine community health programs which is why the DHIS2 integration is critical. From a sustainability standpoint, grounding this work in FHIR Implementation Guides aligns with how countries are increasingly designing their digital health architectures.

3. Organization Expertise and Experience

Ona and Crosscut are bidding on this assignment as a consortium, with Ona serving as the prime contractor. Ona will hold the contractual relationship with UNICEF and bear primary responsibility for all deliverables, project management, stakeholder coordination, and reporting. Crosscut will serve as a subcontractor to Ona, contributing its geospatial microplanning expertise, DHIS2 integration capabilities, and NTD campaign experience to the technical delivery of the ICR.

3.1 Organization Profiles

Ona Systems

Ona is a technology solutions company that uses data to address large, intractable problems such as equitable access to basic healthcare services. Founded in 2013 and headquartered in Burlington, VT, Ona is the creator and technical steward of OpenSRP, a global good digital health platform that has been adopted at national scale in several countries. Ona partnered with the World Health Organization and Google to develop a next-generation version of OpenSRP built natively on HL7 FHIR and designed to support WHO's SMART Guidelines. Ona has also played a key contributing role in the creation of several other global goods, including Google's Open Health Stack and Reveal, a precision health delivery platform.

Ona has successfully delivered hundreds of projects in over 30 countries with the world's leading development agencies and practitioners. As an organization, Ona has a proven track record of working with and supporting UN partners. Ona holds active LTAs with UNICEF (and previously with WHO) and has successfully completed close to 100 contracts across the UN system, including with UNICEF, WHO, WFP, UNDP, UNEP, UNOPS, and UN Women. Ona also provides enterprise data solutions adopted corporately by UN agencies, including UNICEF's Inform platform and WFP's MoDa mobile data collection system.

Crosscut

Crosscut is a certified U.S. small business founded in 2017 and headquartered in Washington, DC, with a mission to unlock data to make global health programs more efficient and effective. The company specializes in geospatial microplanning technology for public health campaigns, operating across Sub-Saharan Africa with a platform now available in 43 countries and three languages.

Crosscut's team of five full-time engineers, data scientists, and global health professionals has worked together for years under a disciplined agile development model with rigorous quarterly planning, two-week sprints, daily standups, and formal testing and code review. The company maintains a broad network of geospatial health analytics specialists and can expand capacity on short notice when project demands require it.

Since 2020, Crosscut has implemented projects across more than 10 countries in geospatial microplanning, immunization supply chain analytics, and NTD campaign support. Clients and funders have included UNICEF, WHO AFRO/ESPEN, USAID/PMI, the Gates Foundation, VillageReach, The Carter Center, Sightsavers, Abt Associates, Amentum, Task Force for Global Health, and the IDA Foundation. Crosscut's CEO, Coite Manuel, brings more than 15 years in

global health supply chain management, including prior service as Senior Supply Chain Advisor in USAID's Global Health Office.

Crosscut's core platform, the Crosscut App (app.crosscut.io), is a freely available, registered Digital Public Good that delivers hyperlocal geospatial maps to support health programming and analysis at sub-district and 100m² granularity or lower across Sub-Saharan Africa and other geographies. The app powers the WHO ESPEN Geospatial Microplanner and holds formal status as a DHIS2 Tier 2 Strategic Technology Partner (Digital Public Good) under a signed agreement with the HISP Centre at the University of Oslo.

3.2 WHO SMART Guidelines L3 Experience

Ona has extensive experience designing and implementing WHO SMART Guideline-based L3 applications, having been involved in conceiving and developing the approach from its earliest stages. This work began when Ona built the first ANC digital health reference application for WHO, prior to the formal launch of the SMART Guidelines initiative. Ona then partnered with WHO and Google to create Open Health Stack, the first reference implementation of the SMART Guidelines framework, which established the foundational architecture for translating L2 clinical recommendations into executable L3 software. From there, Ona developed the first SMART Guidelines L3 application for COVID-19 and subsequently worked with WHO to build a SMART Guidelines-compliant immunization reference application, which was later adapted in partnership with PATH into a global immunization reference product suite. Ona has also led country-level SMART Guidelines adaptation work, including developing a base SMART Guidelines FHIR Implementation Guide for Peru and training the Ministry of Health to customize it for national-level use. In addition, Ona has built SMART Guidelines-informed FHIR-native applications across multiple countries, including a community health worker application now deployed by the Ministry of Health in Uganda with support from UNICEF. While not all of these implementations constitute full published IGs, each represents a complete FHIR-based community health implementation grounded in SMART Guidelines methodology.

OpenSRP 2, Ona's FHIR-native platform, was purpose-built to serve as the L3 execution layer for WHO SMART Guidelines content. Built on Google's Android FHIR SDK as part of the Open Health Stack collaboration, OpenSRP 2 uses HL7 FHIR resources, Structured Data Capture (SDC), and Clinical Quality Language (CQL) to allow subject matter experts to define clinical workflows, decision-support logic, and data capture forms through dynamic configuration rather than code changes. This design enables countries to take WHO's machine-readable L2 guideline content and rapidly adapt it into functional, deployable L3 applications.

Ona's team is deeply embedded in the HL7 FHIR community. Team members have participated in multiple HL7 FHIR DevDays conferences, FHIR training sessions led by the HL7 team, and connectathons. This sustained engagement has led to direct contributions to the FHIR specification itself, including work on how to represent and manage households in the context of primary healthcare delivery in FHIR, as well as contributions to the authoring and implementation of FHIR Implementation Guides. In 2024, Ona worked with the Ministry of Health of Peru (MINSA) in collaboration with PAHO to train MINSA staff on FHIR and support the authoring of a Immunization FHIR Implementation Guide tailored to Peru's national health system. As part of this engagement, Ona helped develop a FHIR-native application for initial pilot testing in the country. This work has had a lasting institutional impact: MINSA has since implemented HAPI FHIR and begun adopting FHIR internally as a foundation for health data exchange across the

organization. Across these efforts, Ona has delivered SMART Guidelines L3 implementations spanning multiple countries, clinical domains, and partner organizations — a depth of applied L3 experience that is, to our knowledge, unmatched in the FHIR community.

3.3 HL7 FHIR Implementation Guide Experience

We have experience authoring and publishing IGs as required in the TOR. Members of our team including Matt Berg and Peter Lubell-Doughtie have been authors on published IGs. Our work in helping to initiate the WHO SMART Guidelines made us very familiar in both the FHIR IG authoring and consuming (L3) process. This includes using FSH/SUSHI to profile, create StructureDefinitions, ValueSets, etc. See Appendix 3 where we provide links to a published IG which our team contributed to authoring with the requested validation criteria.

3.4 Digital Campaign Implementation Experience

Ona and Crosscut bring deep, complementary experience supporting digital implementation of public health campaigns across immunization, NTD mass drug administration, and malaria vector control, with direct engagement at national and subnational government levels across multiple countries and partners.

Malaria IRS and ITN Campaigns

Ona developed with Akros mSpray, one of the earliest digital tools purpose-built for indoor residual spraying campaigns, and supported its use for multiple years in Zambia at national scale, covering over one million structures. Published research demonstrated that IRS campaigns using mSpray achieved statistically significant reductions in malaria incidence compared to IRS without it, validating the impact of geospatial guidance on campaign effectiveness. mSpray was subsequently expanded and rebranded as Reveal, an open-source, Digital Square-approved Global Good that uses spatial intelligence to guide and track delivery of household-level health interventions. In partnership with CHAI, Reveal was extended to additional countries across sub-Saharan Africa for IRS, ITN distribution, and other campaign use cases.

Crosscut was selected as a resource partner on USAID's PMI Evolve Project (prime contractor Abt Associates), a five-year program spanning 21 countries. In the first year, Crosscut conducted design sessions and user testing in Uganda and Ghana for IRS supply estimation, with plans to extend to 10 countries and expand functionality for ITN distribution planning. Crosscut also provided ML-based building classification (85%+ accuracy for identifying spray-eligible structures) to generate printable PDF operational maps for IRS campaigns.

Immunization Campaigns

Ona worked with UNICEF to develop RT-VaMA (Real-time Vaccination Monitoring and Analysis), a digital monitoring tool that enables daily tracking of vaccine coverage, utilization, and wastage during immunization campaigns. Built on ODK technology, RT-VaMA allows health workers to collect data on mobile devices — including in offline settings — and surfaces that data through visualization dashboards that enable decision-makers to identify bottlenecks and course-correct in real time. RT-VaMA was first piloted with over 1,500 health workers across the Philippines during a polio outbreak response and has since been deployed in Papua New Guinea, where it supported a measles and rubella campaign targeting 1.3 million children across 22 provinces.

UNICEF has adopted RT-VaMA as a core component of its immunization innovation portfolio and has developed a toolkit to support its adoption in additional countries.

More broadly, Ona supports UNICEF and WHO's immunization and polio programs through Ona Data, Ona's mobile data collection platform, which serves as a primary tool for campaign data collection and monitoring across Africa. Through this sustained engagement, Ona has developed deep operational familiarity with immunization campaign data flows, including coverage tracking, supply chain visibility, and integration with national reporting systems. Ona has also implemented OpenSRP as an electronic immunization registry in Zambia, Mauritania, and Tunisia, and is currently working with WHO to develop a global reference EIR application based on the SMART Guidelines.

Crosscut has worked directly with UNICEF on digitized electronic microplanning strategies for immunization campaigns in The Gambia, supporting the government in making evidence-based platform decisions. The Crosscut App explicitly supports supplemental immunization activity (SIA) planning, vaccination post identification, session planning, and population-benchmarked coverage estimation. Crosscut was also selected as a Gavi global partner in supply chain analytics (2022, prequalified in Ethiopia), supporting vaccine supply chain visibility through integration with UNICEF's Stock Management Tool (SMT), which is in active use in 20+ countries. This integration enables batch-level visibility from health facility to central medical store to support expiry prevention and stockout detection. Crosscut has since been selected (with its prime partner VillageReach) by Gavi as part of the 6.0 Technical Assistance framework for geospatial microplanning.

NTD Mass Drug Administration (MDA)

In Nigeria, Crosscut partnered with Sightsavers and the Kogi State Ministry of Health to support planning for a schistosomiasis MDA campaign across Lokoja, Kogi, and Ibaji LGAs, targeting 152,724 children aged 5–14. Crosscut provided accessibility analysis, supervisory area delineation, and a CommCare integration for real-time supervision monitoring during the campaign. Crosscut has also integrated with Open Data Kit (ODK) for NTD MDA supervision data management, with The Carter Center's Nigeria program as a named deployment. In addition, Crosscut has developed a CommCare integration for MDA supervision deployed with Sightsavers across Liberia, Guinea-Bissau, and Nigeria, enabling field supervisors to log visits through CommCare while program managers see coverage and supervision status overlaid on catchment maps.

In Liberia, Crosscut piloted the ESPEN Geospatial Microplanner across three counties (Maryland, Grand Cape Mount, and Montserrado) in partnership with Sightsavers to prepare catchment-based microplans for an upcoming NTD MDA campaign. In Haiti, Crosscut collaborated with The Carter Center on MDA campaigns targeting lymphatic filariasis, where Crosscut's population estimates directly informed the decision to shift from fixed-point to door-to-door drug distribution in specific areas. In 2026, Crosscut is expanding NTD program microplanning support to Nigeria, Senegal, Madagascar, Guinea-Bissau, Burundi, Tanzania, Ghana, and Chad.

Understanding of Campaign Data Flows

Across these engagements, the team brings direct operational experience with the full spectrum of campaign data flows. Ona's platforms produce and consume FHIR-based health data and integrate with DHIS2 and other national HMIS. Crosscut's platform outputs include population-benchmarked coverage estimates, supply plan tables, and catchment-area boundaries exportable as GeoJSON and ingestible directly into DHIS2. Crosscut integrates with WHO ESPEN for MDA-specific workflows and DHIS2 via a signed MOU and DHIS2 App Hub app. Together, the team has hands-on experience with coverage tracking, campaign deduplication, Gavi/COVAX reporting requirements, and the DHIS2/OpenSRP ecosystem, the exact data flows that the Integrated Campaign Registry is designed to standardize and connect.

3.7 Geospatial Microplanning Experience

Crosscut's core business is geospatial microplanning, and its technology powers the WHO AFRO ESPEN Geospatial Microplanner, the primary sub-district microplanning tool for NTD preventive chemotherapy programs across 43 African countries.

WHO ESPEN Geospatial Microplanner

The ESPEN Geospatial Microplanner is a white-label version of the Crosscut App, built in partnership with WHO AFRO's ESPEN program and funded by the Gates Foundation. It supports microplanning for NTD MDA campaigns across all WHO AFRO countries. Crosscut also built the ESPEN Schisto Mapper Tool, piloted in Senegal, which transforms schistosomiasis survey and environmental data into actionable sub-district treatment maps within the ESPEN Portal.

In December 2025, Crosscut participated in a WHO workshop in Brazzaville with representatives from 13 countries to demonstrate the ESPEN Microplanner and gather country feedback on microplanning workflows. Earlier in 2025, Crosscut conducted a landscape survey of 56 NTD program leaders across 27 WHO Africa region countries, including 13 in-depth interviews, to understand microplanning practice gaps. The survey found that 82% of respondents engaged in microplanning, but only 46% estimated populations at the community or sub-district level, and 97% relied on Excel with no geospatial tools.

GIS-Based Household Enumeration and Settlement-Level Mapping

The Crosscut App integrates building-level data from Overture and Google Open Buildings (with configurable confidence thresholds) alongside population data from WorldPop, Meta Data for Good, GRID3, and Kontur. The platform pre-generates geographic "land blocks," which are sub-settlement polygons that adapt to local geography rather than rigid administrative grids, enabling health planners to delineate boundaries that reflect actual community structures whether or not they align with formal administrative boundaries. Building classification outputs are compatible with IRS campaign planning, including identifying spray-eligible versus non-spray structures at greater than 85% accuracy via ML.

Key Microplanning Capabilities Deployed In-Country

- Automated catchment area generation from geocoded health facility locations
- Paintbrush-style boundary editing with real-time population updates, requiring no GIS expertise or desktop GIS software
- Fair Supervisory Areas algorithm that automatically divides administrative units into equitable supervisor territories based on population density, geographic barriers, travel time, and configurable supervisor-to-CDD ratios, with support for constrained calculation under fixed budgets
- Travel time heatmaps (walking and driving) with configurable time-band thresholds
- Microplan Collector that allows national planners to disseminate Excel templates to districts, ingest returned files at scale, and visualize aggregated data with population benchmarking

Country Deployments

Countries with named NTD geospatial microplanning deployments include Nigeria, Senegal, Guinea-Bissau, and Liberia, with several additional countries planned for 2026. In 2023, over 50 Ministry of Health staff across multiple countries were trained to independently generate catchment areas and microplans within DHIS2 as part of the DHIS2 Academy.

3.8 Digital Public Good (DPG) Contributions

OpenSRP 2 OpenSRP is an open-source digital health platform in active development by Ona for over 10 years. OpenSRP 2, released in 2022, combines WHO health workflows with the HL7 FHIR data exchange standard to transform how healthcare is delivered and managed at the facility and community level. Core functionalities include client registration and management with automated scheduling and reminders, complete offline functionality with online FHIR server synchronization, in-app reporting, multi-language support, role-based access control, and interoperability with DHIS2, OpenMRS, and other national health information systems.

- DPG Registry: <https://www.digitalpublicgoods.net/r/opensrp>
- Ona's role: Lead developer and maintainer

Open Health Stack OpenSRP 2 is built on Open Health Stack, an initiative Ona contributed to alongside Google and WHO. OHS provides a suite of open-source tools including a FHIR-native Android SDK, Info Gateway, and FHIR Data Pipes tooling. OHS is a registered global good.

- Registry: <https://developers.google.com/open-health-stack>
- Ona's role: Co-developer and maintainer

Crosscut Platform Crosscut is a verified Digital Public Good registered with the Digital Public Goods Alliance. The registered DPG covers the open datasets produced by the Crosscut App: catchment area maps, microplan tables with population estimates, and supply plan tables, all published under a CC0 1.0 license. Outputs are also published on the Humanitarian Data Exchange (HDX). All 11 DPG Standard indicators have been met. Self-reported organizational users include the Ministry of Health EPI program in The Gambia, WHO, and more than 30 MoH users who have incorporated the open datasets into DHIS2 and other national health information systems. Crosscut's open-source footprint extends beyond the dataset DPG to include a publicly

available DHIS2 integration app, six additional public repositories on GitHub (github.com/crosscutio), and Field-Kit, an open-source geospatial data preparation toolkit released in March 2026.

- DPG Registry: <https://www.digitalpublicgoods.net/r/crosscut> (DPG ID: GID0090906)
- Crosscut's role: Lead developer and maintainer

Reveal Reveal is an open-source platform that uses spatial intelligence to guide and track delivery of household-level health interventions, including case detection, contact tracing, and outbreak management. It supports real-time data collection, analysis, and reporting to enable rapid response to public health threats. Ona contributed to the initial development and field implementation of the platform in partnership with Akros.

- Registry: <https://revealprecision.com/>
- Ona's role: Co-developer (with Akros)

3.9 Production Data Integration Experience

DHIS2

Ona has extensive production DHIS2 integration experience across multiple country deployments. This includes leading the integration of Uganda's FHIR data warehouse, which OpenSRP 2 feeds into, with the Ministry of Health's DHIS2 instance, a project that included building MoH capacity to adapt, manage, and maintain the integration independently. Ona also developed a packaged DHIS2 immunization reporting module as part of its work with PATH on the global immunization product suite, and integrated the Zambia Electronic Immunization Registry (ZEIR) platform with the Zambia MoH's DHIS2 system.

Crosscut holds a signed Technology Partner Agreement with the HISP Centre at the University of Oslo, establishing Crosscut as a DHIS2 Tier 2 Strategic Technology Partner (Digital Public Good), a formal designation for DPG-aligned technologies recognized as strategic ecosystem collaborators. The agreement covers API-level integration, joint reference architectures, coordinated country deployments, and joint engagement with donors and governments. Under this partnership, Crosscut has built and maintains a dedicated DHIS2 Microplanning app available on the DHIS2 App Hub. The app enables national DHIS2 administrators to create sub-district catchment areas directly within their DHIS2 instance without needing GIS expertise, connecting to Crosscut's geospatial engine via read-only token authentication and GeoJSON-formatted API endpoints. DHIS2 v41 (May 2024) includes native support for Crosscut catchment visualization with linked data tables. Crosscut was a finalist for DHIS2 App of the Year in 2022, and more than 30 Ministry of Health users have incorporated Crosscut data into DHIS2 instances across Sub-Saharan Africa.

ODK

Ona has deep roots in the ODK ecosystem. Ona developed the XLSForm standard, which is now the dominant form authoring format across ODK, KoboToolbox, and other data collection platforms. Ona also developed Ona Data, a mobile data collection and management platform that powers UNICEF's Inform platform and is used by WHO AFRO to support immunization and NTD campaign programs across Africa.

Crosscut has integrated with ODK for NTD MDA supervision data management, with The Carter Center's Nigeria program as a named production deployment. ODK-based supervision forms are used by field supervisors to record visit data during campaigns, which Crosscut ingests and renders geographically over catchment maps, enabling program managers to track field team coverage in real time.

CommCare

Crosscut has developed and deployed a CommCare integration for MDA supervision monitoring across three countries with Sightsavers: Liberia, Guinea-Bissau, and Nigeria. Field supervisors log visit data through CommCare, and Crosscut connects to the CommCare API to surface supervision coverage alongside catchment boundaries for geographically-informed campaign monitoring. This integration is structurally similar to the ICR's requirement for data connectors between campaign tools and a central registry.

WHO ESPEN Portal

Crosscut powers the ESPEN Geospatial Microplanner and the Schisto Mapper Tool in production across 43 African countries. These are live, Ministry-of-Health-facing tools integrated into the ESPEN Portal, not prototypes or demonstrations.

Data Flows in Low-Connectivity Environments

Both Ona and Crosscut's platforms are designed for low-bandwidth and offline-first conditions. OpenSRP 2 provides complete offline functionality at the point of service, synchronizing with the FHIR server when connectivity becomes available. The Crosscut App supports pre-generated microplan distribution via Excel or PDF for fully offline use, and both its CommCare and ODK integrations use asynchronous sync patterns appropriate for field environments with intermittent connectivity. The Microplan Collector workflow aggregates district-level Excel submissions nationally without requiring internet access at the field level.

Export and Interoperability

All Crosscut outputs are exportable as GeoJSON and compatible with QGIS, ArcGIS, ODK-based tools, and DHIS2 organization unit hierarchies. Supply plan tables are structured for compatibility with UNICEF's Stock Management Tool (SMT) and WHO Joint Application Package (JAP) reporting formats. Ona's platforms export data via standardized FHIR APIs and support integration with national HMIS through established data pipelines.

3.10 Multi-Country Government Engagement

Ona

Ona has supported UNICEF and partners across over 100 projects in more than 30 countries spanning Sub-Saharan Africa and Southeast Asia. This work has involved direct engagement with Ministries of Health, coordination with UN agencies and implementing partners, and sustained programme management across diverse country contexts. Named Ministry of Health engagements include Uganda, where Ona leads the development and integration of the national community health worker platform with UNICEF support; Zambia and Malawi, where Ona

implemented electronic immunization registries integrated with national DHIS2 systems; Liberia, where OpenSRP was selected for national community health use; Tunisia and Mauritania, where Ona deployed electronic immunization registries; Peru, where Ona worked with MINSA and PAHO on FHIR adoption and capacity building; and Papua New Guinea, where Ona partnered with UNICEF, WHO, and the National Department of Health on the RT-VaMA immunization monitoring initiative. Ona also maintains ongoing operational relationships with UNICEF, WHO, PATH, and Gavi through its work on OpenSRP, Ona Data, and the global immunization product suite.

Crosscut

Crosscut has direct engagement with Ministries of Health in at least 10 countries, coordinating across WHO, UNICEF, USAID, Gavi, and implementing partners simultaneously in several of these.

Direct Ministry of Health engagements include:

- Nigeria: Kogi State Ministry of Health, schistosomiasis MDA microplanning
- Ethiopia: Ministry of Health (via UNICEF), supply chain design
- Uganda: Ministry of Health (via PMI Evolve), malaria IRS
- Ghana: Ministry of Health (direct and via PMI Evolve), malaria IRS, CHPS zone mapping
- Liberia: Ministry of Health, ESPEN Microplanner pilot, NTD MDA
- Guinea-Bissau: Ministry of Health, app training and deployment
- Burundi: Ministry of Health, onchocerciasis endemicity mapping
- The Gambia: Ministry of Health EPI, supply chain design, microplanning, and catchment area mapping
- Haiti: Ministry of Health (via The Carter Center), lymphatic filariasis MDA
- DRC: Ministry of Health (via Unlimit Health), schistosomiasis boundary delineation

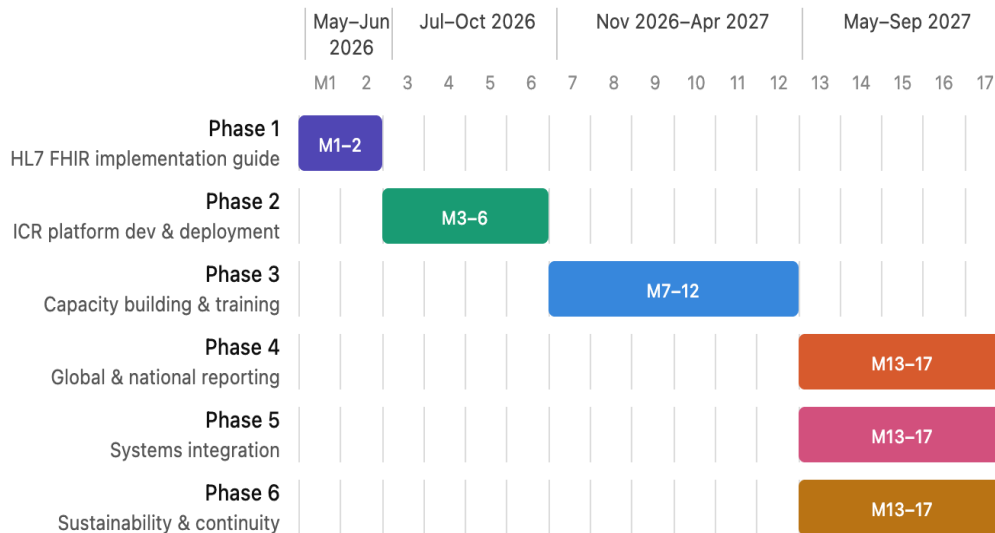
Crosscut has active relationships with UNICEF, WHO AFRO/ESPEN, Gavi (global partner in geospatial microplanning and supply chain analytics, 2022 and 2025), and USAID/PMI (resource partner on PMI Evolve). Crosscut's current Gates Foundation-funded project requires simultaneous coordination across multiple African countries' NTD programs, WHO AFRO's ESPEN secretariat, and country-level Ministry of Health and implementing partner users, all while maintaining a continuously deployed production application. This is the same model of multi-stakeholder management required for the ICR assignment.

French Language Capacity

The team has strong French-language capacity across both organizations. Matt Berg (Ona) has lived and worked in Mali and Senegal and has led technical trainings, workshops, and consulting engagements in French, including for UNICEF Senegal. Crosscut has a French-speaking team member able to lead capacity-building workshops in French and has conducted training in Francophone African countries including Guinea-Bissau and the DRC. The Crosscut App and its documentation are available in French. Together, the team is fully equipped to deliver all aspects of the assignment in French.

4. Implementation methodology/approach and workplan

The following is our proposed approach to implement the project in 6 phases of a span of 17 months which Ona and Crosscut will work together to implement for UNICEF.



Project Governance

Effective delivery depends on close coordination between Ona and UNICEF. We will follow an agile development process, delivering work in iterations that give UNICEF and partners regular opportunities to provide feedback and course-correct as the project evolves. Our proposed cadence includes:

- Weekly check-ins with the UNICEF project contact and relevant HQ staff (e.g., Sean Blaschke)
- Bi-weekly sprint demos during active development phases
- Quarterly steering committee reviews with broader stakeholder participation
- Defined decision gates at the completion of each phase before proceeding

We request that UNICEF designate a primary point of contact responsible for ensuring timely feedback, facilitating access to subject matter experts, and managing other key dependencies on the UNICEF side.

4.1 Phase 1: Development of HL7 FHIR Implementation Guide (Months 1–2)

4.1.1 Approach

Sections 2.1 and 2.2 describe our intended approach to develop and publish the ICR IG for campaign data which will be implemented in this phase.

Stage 1: Inception and Campaign Data Review (Weeks 1–3).

- Inception - The project will start off with a set of project inception calls with the UNICEF team. From this we will develop a project work plan for UNICEF's feedback and sign-off.
- Data review - UNICEF will facilitate access to data from 3–4 existing campaigns which may include NTD mass drug administration (SCH/STH), immunization, malaria, etc. We will review each campaign dataset, cataloguing every data element captured to identify and map the commonalities and divergences against the campaign types to come up with a canonical data model to represent campaigns that we will build into the IG.
- Data components and terminology - We will begin to define which data components must be captured in real-time during active campaigns (doses administered, locations visited, progress against targets) versus reconciled at campaign close (stock counts, final coverage calculations, data quality reviews, JAP-aligned reporting). We will also begin to map out the terminology ValueSets for this

Stage 2: Draft IG Authoring (Weeks 3–6).

In this phase we will develop the v0.1 draft IG following a process detailed in Section 2.2. Key activities include:

- Profile FHIR R4 resources around the CarePlan-based architecture, define ValueSets for campaign types and drug/vaccine products (eg. CVX, WHO ATC, ConceptMaps for local code alignment), and extend the Location resource for GeoJSON boundaries and multi-identifier systems (P-codes, Overture GERS IDs, national facility codes).
- Compile the draft via FSH/SUSHI and the HL7 IG Publisher, with continuous integration configured to validate every change automatically.
- Perform initial data conformance testing by converting real campaign datasets into the proposed FHIR model, identifying where the model fits and where it needs adjustment.
- Circulate the draft as a versioned, browsable publication so reviewers — MoH, UNICEF, WHO, ESPEN, and implementing partners — can inspect actual profiles and examples rather than static documents. Track all feedback in GitHub.

Stage 3: Revision and Endorsement (Weeks 6–8). We will incorporate stakeholder and community feedback, resolve issues identified during data conformance testing, and produce the revised IG endorsed for pilot use.

4.1.2 Deliverables

- Project workplan developed and validated with UNICEF, including the stakeholder engagement plan for IG development and review.
- Draft HL7 FHIR IG for campaign data published as a versioned browsable IG with a public GitHub repository.

- Documented definition of data elements and reporting requirements, including specification of components to be reported in real-time versus reconciled at campaign close, with rationale for each classification.
- Revised HL7 FHIR IG incorporating stakeholder and community feedback, endorsed for pilot deployment, with changelog documenting all changes from the initial draft.

4.1.3 Key Dependencies

This phase produces the IG that serves as the DNA of the entire ICR system. Any delay here cascades directly into Phase 2 development.

- **UNICEF** must facilitate timely access to campaign data from 3–4 campaigns. If data access is delayed, the 8-week window compresses and either the depth of the data review or the stakeholder feedback cycle will be affected.
- **Stakeholder availability** for structured IG review within a two-week feedback window (Weeks 6–8). This requires coordination across MoH, WHO, ESPEN, and implementing partners across multiple time zones.
- **Campaign data quality.** The data provided must be of sufficient quality and completeness to inform meaningful FHIR profiles. If data is incomplete or heavily aggregated, we may need to supplement with additional campaign datasets.

4.2 Phase 2: ICR Platform Development and Deployment (Months 3–6)

4.2.1 Approach

Building on the reference solution architecture described in Section 2.3, this phase develops, deploys, and validates the ICR in two pilot countries through an iterative, sprint-based process. Development and in-country engagement are tightly coupled: the optional in-country trip for each pilot (up to 10 days, subject to UNICEF approval) will serve double duty, supporting both prototype testing and initial hands-on training with MoH and implementing partners.

Months 3–4: First Pilot Country – Development and Deployment (Sprints 1–4)

Development will follow two-week Agile sprints with planning, development, testing, review, and stakeholder demo at each cycle.

Sprint 1: Infrastructure and Core FHIR Store. We will stand up the HealthCare API server in a hosting cloud hosting environment agreed with UNICEF during inception, configured with the IG profiles, extensions, and terminology from Phase 1. We will deploy Cinder alongside the FHIR store for data browsing, validation, and quality management.

Note: To meet the tight timelines and security requirements for the project cloud hosting will likely be required for the initial pilot testing. After the pilot phase, we will work with UNICEF and the MoH to explore what in-country hosting solutions could look like if required by their preferences and requirements. These are details that are unknown now that must be explored and figured out during the project to ensure they align with expectations, available resourcing, etc.

Sprint 2–3: Campaign Integration Connectors

For the two campaigns identified by the UNICEF Country Office, we will build OpenFn connectors that extract data from whatever collection tools those campaigns use (ODK, DHIS2 Tracker, CommCare, or others), transform it into FHIR-conformant resources per the IG, and test loading data it into the FHIR store. Each connector will follow the transformation layer design described in Section 2.3. Critically, the campaigns continue using their existing data collection tools – the ICR meets countries where they are rather than requiring new tools. During these sprints we will also configure Cinder's deduplication and data quality functions for the specific data types flowing from each campaign, as cross-campaign deduplication of households and locations will be essential to registry integrity.

Sprint 4: In-Country Testing, Validation, and Training

We will travel to the first selected country for initial piloting of the ICR approach. During this trip, we will conduct end-to-end testing with actual campaign data alongside MoH and UNICEF Country Office staff, validate cross-campaign data reuse (e.g., household locations collected by one campaign available to the next), and document integration workflows. The trip will also serve as the primary opportunity for hands-on training with MoH data managers, campaign coordinators, and IT staff in this country (see Phase 3). We will work directly with MoH counterparts to walk through ICR operations, conduct UAT to identify usability issues, and gather feedback that will inform the v2 update. This combined testing-and-training approach ensures that MoH staff are engaged from the moment the system is live, not trained on it months later as a separate exercise.

Month 5: Pilot Feedback and IG/System Update (v2)

Based on feedback from the first pilot deployment, we will update the IG and ICR system to produce a v2 version. This update cycle will address issues identified during live testing – data model gaps, integration edge cases, performance issues and usability feedback from MoH and UNICEF staff. All changes will be documented in the IG and ICR reference system change-logs

Month 6: Second Country Deployment

We will deploy the updated ICR (v2) in the second pilot country, adapting the system for the local context. This includes configuring connectors for the two campaigns identified by that country's UNICEF Country Office, adapting ValueSets and ConceptMaps for country-specific terminology, and configuring the Location hierarchy for the country's administrative structure. As with the first country, the optional in-country trip (up to 10 total days) will combine prototype testing with hands-on training and stakeholder engagement, onboarding MoH and partner staff while the system is being validated. The second deployment will test the portability of the reference solution and surface any country-specific adaptations needed for the replication toolkit.

Training will be conducted in both English and French as required by the pilot country context. For Francophone countries, Team Lead Matt Berg, who grew up and worked extensively in French West Africa and/or

Clara Burgert, a bilingual epidemiologist with extensive experience supporting Francophone country programs will support the in-country piloting and training.

4.2.2 ICR Solution Package

As part of this phase, we will complete the initial implementation of the ICR solution documented in Section 2.3. This is minus some of the later phase integrations like the data warehouse and DHIS2 integrations. The software will be developed in an agile methodology with code being made available in Github. The ICR will be built using open-source technologies with any new software developed made available under Apache 2.0 or equivalent open source license.

4.2.3 Deliverables

- Prototype ICR system built and tested in the first pilot country, including FHIR store, data browsing interface (Cinder), and transformation connectors for two campaigns.
- Documented workflows for the integration of two campaigns in the first pilot country, including data flow diagrams, connector configurations, and data quality management procedures.
- Updated IG and ICR Solution package system (v2) incorporating pilot feedback.
- Second-country deployment with at least two campaigns integrated, including country-specific adaptations documented.
- Packaged ICR solution with: user manual and administration guide, technical documentation for replication, security and data protection documentation, and open-source licensing framework (Apache 2.0).

4.2.4 Key Dependencies

- **Finalized IG from Phase 1** (deliverable d). Platform development cannot begin until the IG is endorsed for pilot use.
- **UNICEF Country Offices** must identify the two campaigns for integration in each pilot country and facilitate MoH engagement before Month 3. Late campaign identification will compress development sprints.
- **Campaign schedules** in pilot countries must overlap with the project timeline. If no campaign runs during Months 3–6, there will be no live data to integrate and validate, limiting testing to historical data.
- **API availability** from country systems (DHIS2, ODK, etc.) is dependent on country system readiness. UNICEF does not guarantee API availability from government systems. Where APIs are unavailable, we will use file-based or bulk-export integration approaches.
- **In-country travel** (up to one trip per pilot country, maximum 10 days) requires prior written UNICEF approval. Travel timing affects our ability to conduct on-the-ground testing and training.

4.3 Phase 3: Capacity Building and Training (Months 7–12)

Note: Due to travel limitations, the in-country training will be done in Phase 2 during the country site visits when the system is being tested with live data and MoH staff are directly engaged. Phase 3 will therefore focus on formalizing, documenting, and extending what was learned during the pilots into documentation and improved training materials.

4.3.1 Approach

Documentation, Job Aids, and SOPs (Months 7–12)

Following the pilot deployments and in-country training (in Phase 2), we will develop SOPs, job-aids and training materials that will enable UNICEF and MoH staff to operate the ICR independently with minimal support and train additional resources.

The training / documentation materials will be made available online format and will include:

- Step-by-step job aids for each user role, grounded in the actual workflows established during the pilots
- Standard operating procedures (SOPs) for integrating ICR data across current and future campaigns
- Visual workflows showing how data moves between campaign tools, the ICR, DHIS2, and reporting outputs
- Troubleshooting guides for common issues encountered during the pilots

Integration with Microplanning Workflows

We will work to provide guidance to UNICEF DAPM teams leading the microplanning to ensure they know how to access the ICR data for microplanning purposes. The Crosscut platform, which is part of the ICR package, will be made available for microplanning in this phase if desired.

4.3.2 Deliverables

- Training and/or Training-of-Trainers (ToT) sessions conducted in each pilot country for MoH and implementing partners (delivered during Phase 2 in-country engagements), with documented attendance and curricula delivered.
- Role based guides, standard operating procedures (SOPs) for integrating ICR data across current and future campaigns
- Documented approach and implementation of integration of ICR data into national reporting and microplanning workflows.

4.3.3 Key Dependencies

- **Working ICR system from Phase 2.** Training is delivered during the pilot trips; documentation is developed afterward based on lessons learned. If Phase 2 deployment is delayed or incomplete, both training and documentation quality will be affected.

Phases 4, 5, and 6: Concurrent Workstreams (Months 13–17)

4.4 Phase 4: Global and National Reporting Alignment (Months 13–17)

4.4.1 Approach

Ensure that UNICEF-supported MDA campaigns report using data formats aligned with the WHO Joint Application Package (JAP) and that reporting processes are embedded within national cycles so they survive beyond the project period.

JAP Reporting Support

We will work with WHO and ESPEN focal points to map ICR data elements to specific (JAP) form fields, covering campaign coverage by administrative unit, drug consumption and wastage, demographic breakdowns, and program performance indicators. We will configure the ICR package export modules to generate submission-ready outputs that country teams can validate and submit to WHO/AFRO/ESPEN through established channels. The mapping and export logic will be documented so that future changes to JAP specifications can be accommodated without vendor involvement.

UNICEF DAPM Monitoring Support

We will support the UNICEF DAPM team in accessing ICR data for microplanning and real-time monitoring activities they lead. This will primarily consist of ensuring that DAPM can independently access and use ICR data without ongoing vendor support.

4.4.2 Deliverables

- JAP-aligned data outputs and export functionality developed and validated for UNICEF-supported MDA campaigns, including documented field-level mapping between ICR data elements and JAP form specifications.
- Reporting outputs aligned with JAP forms and validated against WHO and national requirements, with documented procedures for MoH staff to independently generate and submit reports.
- Data access workflows established and documented to enable use of ICR data by UNICEF DAPM for real-time monitoring.

4.4.3 Key Dependencies

- **Operational ICR with real campaign data** from Phases 2–3. Reporting alignment requires actual data flowing through the system.
- **WHO/ESPEN** for current JAP form specifications and validation of the mapping. If WHO updates JAP requirements during the project, this may require a change order.
- **UNICEF DAPM team** availability to define data access needs.

4.5 Phase 5: Systems Integration (Months 13–17)

4.5.1 Approach

During Phase 2, we will have already built the data collection connectors for campaign data needed for country piloting. This phase will thus focus on integrations for DHIS2, the data warehouse and two-way integration and WHO's Geospatial Microplanner (Crosscut Platform).

Analytics Add-On: FHIR-to-Data-Warehouse

The SQL on FHIR layer described in Section 2.3 transforms FHIR resources into flat, queryable tables in a relational data warehouse using ViewDefinition resources that are part of the IG. During this phase, we will put in place the tools to automate these processes to help generate the reporting requirements identified during Phase 3 and 4 and to populate the JAP forms.

DHIS2 Integration

We will develop a connector that will enable importing data from the ICR into DHIS2. This will include importing of location data (organization units), aggregate indicators into the ICR. We will leverage an existing platform like OpenFN to do this vs. developing a bespoke solution. The connector will be openly available, documented and included in the replication toolkit in Phase 6.

Two-Way WHO Geospatial Microplanner Integration

The WHO Geospatial Microplanner is powered by the Crosscut platform (see Section 3.7), meaning this integration is between two components within our consortium's technical control, significantly reducing integration risk. The Crosscut will extend its platform to consume FHIR data from the ICR eg. health facility locations enabling users to define catchment areas, assign population estimates, calculate risk scores, and delineate supervisory zones. Enriched data will be pushed back into the ICR's FHIR store, creating a bidirectional flow that will help bolster the use of ICR data for microplanning.

4.5.2 Deliverables

- ViewDefinition based data warehouse connector developed and documented that is able to meet JAP / country reporting and analytical needs.
- Production-grade data connectors for DHIS2 (bidirectional) and the UNICEF-designated platform (ODK or as confirmed), with documented data flows, error handling, and testing evidence.
- Two-way WHO Geospatial Microplanner integration established and tested, with documented data flows for ICR-to-Microplanner import and Microplanner-to-ICR pushback.

4.5.3 Key Dependencies

- **Stable ICR platform with real campaign data** from Phases 2–3.
- **DHIS2 API availability** from country instances. DHIS2 versions and API capabilities vary by country and are not guaranteed by UNICEF. Where APIs are limited, we will use DHIS2's import/export functionality as a fallback.

- **Crosscut platform readiness** for FHIR data consumption and pushback. As a consortium partner, this dependency is managed internally through joint sprint planning.

4.6 Phase 6: Sustainability and Continuity (Months 13–17)

4.6.1 Approach

Sustainability is not a bolt-on activity at the end of the project – it is a design principle embedded from Phase 1 (open-standard IG, open-source licensing) through Phase 3 (capacity building, MoH ownership orientation). Phase 6 formalizes the sustainability outcomes of these earlier investments, ensures the ICR is institutionalized within national systems, and packages the solution for replication.

Institutionalization within National Reporting Cycles

Working with MoH and UNICEF in each pilot country, we will work to embed ICR reporting processes into MoH systems (eg. via DHIS2 and JAP reporting). We will work with UNICEF to try and ensure that integrated campaign reporting is recognized as a standardized requirement within government systems, not a project-specific activity. This may involve supporting MoH directives or SOPs that formalize ICR-based reporting, integrating ICR data submission into existing HMIS reporting calendars, and documenting the reporting workflow for independent MoH maintenance. The specific mechanisms will vary by country and will be developed in close consultation with MoH leadership and UNICEF Country Offices.

System Handover and Government Ownership

We will execute a structured handover to ensure MoH technical counterparts can maintain and extend the ICR beyond the project period. This includes transferring the source code repository with CI/CD pipeline documentation and build/deployment instructions, delivering a system administration and ICR maintenance guide covering FHIR store management, connector monitoring, data quality procedures, and troubleshooting.

Replication Toolkit

The replication toolkit will package the documentation and developed systems to enable expansion for additional countries. This will include:

- The published IG with instructions with data governance / instructions on how to localize modify by country.
- ICR package including all software components with documentation, supporting deployment scripts, etc.
- Connector templates for common campaign data sources (ODK, DHIS2 Tracker, CommCare) with documented configuration points
- Training materials developed in from Phase 3, packaged for reuse

The toolkit will be hosted in a public GitHub repository under the Apache 2.0 license, ensuring any country, organization, or donor can access, use, and extend it without restriction.

4.6.2 Deliverables

- ICR reporting processes embedded and documented within national MoH reporting cycles in each pilot country, including any supporting MoH directives, SOPs, or calendar integrations.
- System components and maintenance documentation (source code, documentation, maintenance guides) shared with MoH counterparts.
- Replication toolkit for expansion to additional countries, published in a public GitHub repository under Apache 2.0 license.

4.6.3 Key Dependencies

- **Successful Phase 2 and 3 outcomes.** Sustainability is only meaningful if the ICR is operational and MoH staff are trained.
- **MoH institutional commitment.** Embedding the ICR into national reporting cycles requires political will and institutional buy-in that UNICEF can influence but the vendor cannot unilaterally deliver.
- **Lessons from both pilot countries.** The replication toolkit depends on sufficient implementation experience to produce credible, tested guidance.

4.7 Limitations and Exclusions

- Integrations are limited to country-owned systems and globally recognized platforms (DHIS2, OpenSRP, ODK, and other government-approved tools). No integration with UNICEF internal systems is required. Integration with custom, bespoke, or locally developed solutions is not in scope for this phase.
- The ICR does not replace existing data collection tools. Campaigns continue using their existing tools; the ICR provides the interoperability and exchange layer.
- The Vendor is not responsible for procuring or distributing NTD medicines, conducting MDAs, or leading microplanning activities.
- Any development requirements that arise outside the scope of this ToR will be managed through a formal Change Order process, with scope, pricing, and timeline mutually agreed through contract amendment.
- The scope of this engagement covers the initial 17-month period of performance. Maintenance and support services beyond this period are optional and subject to UNICEF's decision to exercise the extension.
- Data ownership rests with country governments. Data sharing and access will be governed by the respective government entity's data-sharing policies. Ona cannot guarantee the availability or shareability of datasets generated during the project.
- The Vendor is not responsible for generating geospatial data (e.g., building footprints, satellite imagery, or population estimates). Where such data is needed, it is expected to be provided by UNICEF, MoH, or sourced from publicly available datasets.

- Deployment is limited to two pilot countries as defined in the ToR. Scale-up beyond these countries is outside the scope of this engagement, though a replication toolkit will be delivered to support future expansion.

7. Project Team

This project is led by Ona Systems, Inc. as Prime/Lead Contractor. Ona holds full contractual accountability to UNICEF for performance of the entire assignment and is responsible for overall technical direction, quality, and timely completion of all deliverables. Crosscut serves as a named consortium partner, contributing specialized capabilities in geospatial microplanning, campaign data analysis, and country engagement.

7.1 Organization Responsibilities

Ona Systems leads the development of the HL7 FHIR Implementation Guide, the ICR reference platform and all data integrations, documentation, standard operating procedures, training materials, and quality assurance. Ona leads in-country technical engagement for the country pilots and coordinates all project reporting to UNICEF.

Crosscut leads geospatial platform adaptation for FHIR data consumption from the ICR, two-way integration with the WHO ESPEN Geospatial Microplanner. Crosscut will provide support in the design of the initial IG, on ground-support for the country piloting and ongoing documentation and replication package development.

7.2 Consortium Structure

This proposal is submitted by Ona Systems, Inc. as Prime/Lead Contractor. Ona holds the relevant Long-Term Agreement (LTAS) with UNICEF and retains full contractual accountability to UNICEF for performance of the entire assignment, including all activities carried out by consortium members. Crosscut serves as a consortium partner for the full duration of the engagement.

The consortium arrangement is additive: Crosscut brings specialized capabilities in geospatial microplanning technology, campaign data analysis, and in-country implementation that complement and strengthen Ona's core competencies in HL7 FHIR standards, digital health platform development, and health system interoperability. Crosscut's scope does not substitute for any of the core competencies required of the Prime/Lead Contractor under this ToR.

Coordination between Ona and Crosscut is managed through joint sprint planning under Ona's two-week Agile development cycle, a shared project management platform for task tracking and issue management, and regular synchronization between Matt Berg (Ona) and Coite Manuel (Crosscut). All project-related Crosscut activities are coordinated through and accountable to Matt Berg as project lead. Decisions affecting scope, timeline, or deliverables are escalated to UNICEF through Ona. Crosscut commits to full compliance with UNICEF's applicable policies and requirements for the duration of the engagement.

A signed letter of commitment by Crosscut is attached in the Appendix.

7.3 Key Personnel

Matt Berg – Project Lead *Organization:* Ona Systems, Inc. *Role on this project:* Project Manager, System Architect, Named FHIR IG Author, In-Country Lead for Côte d'Ivoire *Overall LOE:* 50% across 17 months *Languages:* English, French *CV:* Annex B

Matt Berg is the Chief Executive Officer of Ona Systems and the named FHIR IG author for this assignment. He brings close to 20 years of experience in digital health with deep expertise in HL7 FHIR standards and health data systems. Matt was instrumental in developing the WHO SMART Guidelines framework and has been a leading advocate for FHIR adoption across global health systems, having built and overseen FHIR-native applications across multiple country deployments. He has driven the concept of integrated campaign registries through sustained engagement with UNICEF, WHO, and other global partners. Matt is fluent in French, having grown up and worked extensively in Francophone West Africa, and will serve as the in-country technical lead for the Côte d'Ivoire pilot. His CV, including links to previously authored and publicly inspectable FHIR Implementation Guides, is provided in Annex B.

John Mashuma – Lead Developer and Interoperability Specialist *Organization:* Ona Systems, Inc. *Role on this project:* Lead Developer, SRE, Security and Interoperability Specialist *Overall LOE:* 60% across 17 months *Location:* Nairobi, Kenya *Languages:* English *CV:* Annex B

John Mashuma is a senior software engineer and SRE expert at Ona Systems based in Nairobi, Kenya. He serves as Lead Developer and Interoperability Specialist for this assignment, with primary responsibility for the ICR platform build, FHIR store configuration and management, integration connectors and system hosting and security. John has direct experience developing and maintaining secure software systems that have been implemented at scale. His full CV is provided in Annex B.

Faith Mutua – Documentation, Training and QA Specialist *Organization:* Ona Systems, Inc. *Role on this project:* Documentation, Training and QA Specialist *Overall LOE:* 33% across 17 months *Location:* Nairobi, Kenya *Languages:* English *CV:* Annex B

Faith is a member of the Ona team in Nairobi, holding a BSc in Mathematics and Computer Science with a background in data analysis and implementation support. On this project, she leads the development of standard operating procedures, role-based job aids, and training materials for Ministry of Health and implementing partner staff in both pilot countries, and supports quality assurance across project deliverables. Her work is concentrated in Phase 3 (Months 7–12), building on operational learning from the Phase 2 pilot deployments. Her full CV is provided in Annex B.

Coite Manuel – Senior Advisor *Organization:* Crosscut *Role on this project:* Senior Advisor, Strategic Partnerships; in-country participation for Sierra Leone pilot *Overall LOE:* 5% across 17 months, with higher engagement during Phase 2 *Location:* Washington, DC *Languages:* English, Spanish

Coite Manuel is the Founder and Chief Executive Officer of Crosscut, a verified Digital Public Good and DHIS2 Tier 2 Strategic Technology Partner. He brings more than 20 years of experience in global health supply chain strategy, digital health system design, and multi-country implementation leadership, including five years as Senior Supply Chain Advisor in USAID's Global Health Office. He has led complex multi-country engagements with Ministries of Health, UNICEF, WHO/ESPEN, and USAID/PMI across more than 10 Sub-Saharan African countries across NTD mass drug administration, immunization, and malaria programs. He holds an M.S. in Industrial Engineering from Georgia Institute of Technology and a B.S. in Applied Mathematics and Mathematical Economics from Hampden-Sydney College. On this project, Coite serves as Senior Advisor, providing strategic oversight and supporting stakeholder engagement, and joins the in-country deployment to the Sierra Leone pilot during Phase 2. His full CV is provided in Annex B.

James McKinnon – Senior Business Analyst *Organization:* Crosscut *Role on this project:* Senior Business Analyst, Campaign Data Lead *Overall LOE:* 21% across 17 months *Location:* Atlanta, GA *Languages:* English

James McKinnon is a global health senior data analyst and geospatial data specialist with more than 10 years of experience delivering analytics and field operations across Sub-Saharan Africa, South Asia, and beyond. At Crosscut, he leads data-intensive analytical work including GIS-based operational map production, raster analysis, and network modelling for health campaigns. He has led UNICEF-funded supply chain and immunization assessments in Ethiopia and The Gambia – including the personal geocoding of The Gambia's entire national vaccination network – and served as Head of Supply for Médecins Sans Frontières in South Sudan, where he designed and executed a cold chain strategy for an emergency measles vaccination campaign. He coordinated supply chain data across 38 countries during his time with USAID. On this project, James leads campaign data discovery and analysis in Phase 1, country implementation support for the Sierra Leone pilot in Phase 2, JAP output coordination and NTD reporting workflows in Phase 4, and contributes to systems integration and sustainability activities in Phases 5 and 6. He holds an M.A. in International Development from the Josef Korbel School of International Studies and a B.A. in Mathematics and Philosophy (Summa Cum Laude, Phi Beta Kappa) from Wabash College. His full CV is provided in Annex B.

Clara Burgert – Country Engagement Lead, Sierra Leone *Organization:* Crosscut *Role on this project:* Senior Country Engagement Lead for Sierra Leone pilot *Overall LOE:* 5%, concentrated in Phase 2 *Location:* Silver Spring, MD *Languages:* English (fluent), French (fluent)

Clara Burgert is a PhD-trained infectious disease epidemiologist with more than 20 years of experience applying advanced quantitative and spatial methods to neglected tropical disease programs, immunization campaigns, malaria surveillance, and public health intervention evaluation across more than 20 countries. She holds a PhD in Infectious and Tropical Diseases

Epidemiology from the London School of Hygiene and Tropical Medicine and an MPH in Global Epidemiology from Emory University. She currently leads multi-country evaluations of geospatial microplanning for community drug distribution programs and serves as project lead for Francophone country activities, managing stakeholder engagement with Ministries of Health and implementing partners in French. She has facilitated more than 10 regional trainings in epidemiology, GIS, and data visualization for Ministry of Health and partner staff across multiple countries. On this project, Clara serves as the senior country engagement lead for the Sierra Leone pilot deployment in Phase 2, drawing on her deep NTD program knowledge and MoH engagement experience. Her full CV is provided in Annex B.

Sam Hoogewind – Software Engineer, Crosscut Platform Development *Organization:* Crosscut
Role on this project: Software Engineer, Crosscut platform FHIR adaptation and analytics layer
Overall LOE: 25% across 17 months, concentrated in Phase 5 *Location:* Washington, DC *Languages:* English

Sam Hoogewind is a software engineer at Crosscut specializing in cloud-based geospatial data pipelines, population modelling algorithms, and frontend visualization for global health applications. He engineers scalable AWS data pipelines supporting real-time geospatial data access across 45+ African countries at 99%+ uptime, and designed the foundational population raster algorithm that serves as the core population data layer across Crosscut's platform. He holds a B.S. in Computer Science (Magna Cum Laude, GPA 3.80) from Calvin University. On this project, Sam supports Crosscut platform adaptation for FHIR data consumption in Phase 2 and leads development of the Crosscut platform FHIR integration and analytics layer components in Phase 5. His full CV is provided in Annex B.

Emmanuel Koh – Software Engineer, WHO ESPEN Geospatial Microplanner Integration *Organization:* Crosscut *Role on this project:* Software Engineer, WHO ESPEN Geospatial Microplanner two-way integration *Overall LOE:* 5%, concentrated in Phase 5 *Location:* Boston, MA *Languages:* English

Emmanuel Koh is a full-stack software engineer at Crosscut who owns the WHO ESPEN Schistosomiasis Mapper Tool end-to-end — a publicly deployed platform on WHO's ESPEN Portal supporting NTD preventive chemotherapy campaign planning across 30+ countries and 88,000 administrative boundaries. He coordinated with WHO's ESPEN team to define a dedicated survey data API and established the boundary-matching workflow for 40+ countries, giving him direct institutional knowledge of the ESPEN API and data architecture that the ICR's Phase 5 integration requires. On this project, Emmanuel leads the two-way integration between the ICR and the WHO ESPEN Geospatial Microplanner during Phase 5, drawing on his existing ESPEN API relationships and platform ownership to minimize integration complexity and risk. His full CV is provided in Annex B.

Brianna Poulos – Technical Advisor *Organization:* Crosscut *Role on this project:* Technical Advisor
Overall LOE: 2% *Location:* Washington, DC *Languages:* English

Brianna Poulos is a senior software engineer and technical leader with 15 years of experience designing and operating large-scale cloud-native systems. She leads engineering for Crosscut's geospatial modelling platform, which powers national-scale public health campaign planning across 43 Sub-Saharan African countries including the WHO AFRO ESPEN Geospatial Microplanner. Prior to Crosscut, she spent eight years at the Johns Hopkins University Applied Physics Laboratory as a Senior Staff Engineer on mission-critical secure systems, where she led OpenStack deployment initiatives and authored research on cloud data protection. She holds an M.S. in Computer Science from Johns Hopkins University and a B.S. in Computer Engineering (Valedictorian) from North Carolina State University. On this project, Brianna provides technical advisory and security architecture review, drawing on her expertise in secure cloud system design relevant to UNICEF's information security requirements. Her full CV is provided in Annex B.

Appendix 3

Proof of Compliance – FHIR Implementation Guide Criteria

The following evidence demonstrates that our submission meets the requirements for the IG deliverable.

1. *Published, inspectable FHIR IG*

- Live IG (browsable): <https://ona-health.github.io/smart-immunizations-minsa/>
- GitHub repository (FSH/SUSHI source):
<https://github.com/ona-health/smart-immunizations-minsa>
 - FSH resources -
<https://github.com/ona-health/smart-immunizations-minsa/tree/main/input/fsh>
- Artifact index (profiles, extensions, value sets, code systems, examples):
<https://ona-health.github.io/smart-immunizations-minsa/artifacts.html>
- ImplementationGuide FHIR resource (JSON):
<https://ona-health.github.io/smart-immunizations-minsa/ImplementationGuide-minsa.gov.pe.immunizations.json>

2. *Working FSH/SUSHI profiles (R4)*

This IG is authored in FHIR Shorthand (FHS) and compiled with SUSHI.

- **FHIR version:** R4 (4.0.1) is configured here. [Sushi-config.yaml](#). Note this includes the IG author Matt Berg (team lead on this project).
- **FSH source location:**
<https://github.com/ona-health/smart-immunizations-minsa/tree/main/input/fsh> – 26 FSH files totaling 3 profiles, 3 extensions, 4 value sets, 2 code systems, 7 PlanDefinitions, and 6 Questionnaires.

FSH Source and SUSHI compiled FHIR artifacts

Artifact type	FSH source (GitHub)	Rendered page (live IG)
StructureDefinition (profile)	MINSAImmunization.fsh	StructureDefinition-MINSA.Immunization.html
ValueSet	MINSAVaccineVS.fsh	ValueSet-MINSAVaccineVS.html

Extension

[MINSADepartamento.fsh](#)

[StructureDefinition-MINSADepartamento.html](#)

Compilation evidence

All three required artifact types render cleanly on GitHub Pages – HTML pages exist, snapshots and differentials resolve, base resource references bind. If any FSH file had failed to compile, the corresponding page would not exist on the published site.

3. *Named FHIR IG author on proposed team with verifiable CV*

Matt Berg who is the project lead for Ona for the project is a listed author on this IG that was developed in collaboration with MINSA in Peru as part of the FHIR adoption support Ona provided MINSA through Paho.

Matt Berg can be found on the authors list here:

<https://ona-health.github.io/smart-immunizations-minsa/>

1.5 Autores

- Jose Enrique Perez Lu (jperez@minsa.gob.pe)
- Matt Berg (mberg@ona.io)
- Peter Lubell-Doughtie (peter@ona.io)

And also on the SUSHI config here IG authors can be listed.

<https://github.com/ona-health/smart-immunizations-minsa/blob/main/sushi-config.yaml>

Service Level Agreement (SLA)

Maintenance and Support Services – Integrated Campaign Registry

Note: this is meant to indicate how an SLA for the ICR could be structured. It is impossible at this time (before the project has even started) to understand what the final terms and conditions of the SLA will be. There are many factors that would need to be understood and accounted for before we could develop an SLA that we would ensure would meet the needs of both UNICEF and Ona.

The following provides a very basic example of what an SLA for an Ona managed, cloud hosted version of ICR could look like.

Things it does not include or would need to be accounted for:

- *It does not include a cost for hosting. We do not know what the needs would be in terms of number of countries, campaigns, etc.*
- *If support for in-country hosting is required, the SLA would need to be developed on a per-country basis. If the system is hosted by a MoH, for example, many factors related to the hosting of the system would be completely out of our control.*
- *This does not account for source systems - eg. ODK potentially not working.*
- *It does not account for any additional feature development or capacity building. Those would need to be budgeted scoped exercises.*

If an SLA is required, we would request to UNICEF that development of the SLA is built into the process. This could be added scope in a final phase deliverable.

1. Support Model

The Vendor will provide maintenance and support for all ICR components, including the FHIR store (HealthCare API), data transformation and ingestion connectors (OpenFn), the data browsing and quality management layer (Cinder), the geospatial and microplanning layer (Crosscut), the analytics/data warehouse layer, and DHIS2 integration connectors.

Helpdesk and Ticketing. All support requests will be submitted through a dedicated ticketing system (e.g., GitHub Issues or Jira Service Management). Each ticket will be assigned a unique identifier, a severity level, and an assigned owner. UNICEF and MoH focal points will have direct access to submit, track, and comment on tickets.

Escalation Procedures. Issues not resolved within the target resolution window will be automatically escalated. Level 1 (Support Engineer) handles initial triage and known-issue resolution. Level 2 (Senior Developer) handles complex bugs and configuration issues. Level 3 (Technical Lead / Architect) handles critical system failures, data integrity issues, and architectural decisions. Escalation from L1 to L2 occurs if no resolution is achieved within 50% of the target resolution time. L2 to L3 escalation occurs at 75% of the target resolution time.

2. Response and Resolution Timelines

All response and resolution times are measured in business hours (see Section 4 for hours of operation).

Severity	Definition	Response	Resolution	Example
Critical (P1)	Production system down or data loss affecting all users; no workaround available.	2 business hour	8 business hours	FHIR store unresponsive; data ingestion pipeline failure during active campaign.
High (P2)	Major feature degraded with significant user impact; workaround may exist but is not sustainable.	4 business hours	3 business days	DHIS2 connector failing for one pilot country; deduplication engine producing incorrect matches.
Medium (P3)	Minor feature issue or degradation; workaround available and acceptable for continued operations.	1 business day	5 business days	Dashboard visualization rendering incorrectly; non-critical report formatting error.
Low (P4)	Cosmetic issue, enhancement request, or documentation update with no operational impact.	2 business days	Next scheduled release	UI label correction; documentation clarification.

Response time is defined as acknowledgment of the ticket with an assigned severity and owner. Resolution time is defined as delivery of a fix, workaround, or patch that restores functionality. Permanent fixes may follow in a subsequent release if an interim workaround resolves the immediate impact.

3. Issue Tracking, Reporting, and Resolution Management

- **Ticket Lifecycle:** Each issue follows a standard lifecycle: Open → Triaged → In Progress → Resolved → Closed. Tickets are closed only after confirmation from the reporting party.
- **Monthly Reporting:** The Vendor will deliver a monthly support summary to UNICEF including: total tickets opened and closed, breakdown by severity, mean response and resolution times, SLA compliance rate, and a summary of outstanding issues.
- **Quarterly Review:** The Vendor and UNICEF will conduct a quarterly service review to assess SLA performance, identify systemic issues, and agree on improvement actions or SLA adjustments as needed.

4. Availability and Coverage

- **Hours of Operation:** Monday through Friday, 08:00–17:00 East Africa Time (EAT, UTC+3), excluding public holidays observed in pilot countries.
- **Critical Issue Coverage:** For P1 (Critical) issues occurring during an active campaign window, the Vendor will provide best-effort after-hours response within 4 hours of notification via a designated emergency contact.

- **System Uptime Target:** The Vendor will target 99.5% uptime for cloud-hosted ICR components during business hours, excluding scheduled maintenance windows communicated at least 48 hours in advance.



Geospatial Microplanning and Supply Chain Analytics

crosscut.io
coite@crosscut.io
+1-703-727-8784
Washington, DC | USA

April 14, 2026

To:
UNICEF
Supply Division

Re: Letter of Commitment — Consortium Partner Participation, UNICEF Terms of Reference for the Design, Development, and Deployment of the Integrated Campaign Registry (ICR), Annex B – LOT 1C

Dear Sir/Madam,

Crosscut hereby confirms its participation as a consortium partner to Ona Systems, Inc. ("Ona"), the Prime/Lead Contractor, in Ona's proposal submitted in response to the above-referenced UNICEF Terms of Reference.

Crosscut is available for the full duration of the engagement and commits to fulfilling its agreed responsibilities within the consortium as defined in the proposal.

Crosscut acknowledges that Ona Systems retains full contractual accountability to UNICEF for performance of the entire assignment, and that any changes to consortium membership after contract award require prior written approval from UNICEF.

Yours sincerely,

A handwritten signature in black ink, appearing to read 'Coite Manuel', with a stylized, cursive script.

Coite Manuel

Founder & Chief Executive Officer

Crosscut

April 14, 2026

MATTHEW L. BERG

46 Brewer Parkway
South Burlington, Vermont, 05403, USA
+1.202.823.5864

mlberg@gmail.com

PROFESSIONAL EXPERIENCE

Ona

October 2013-Present

Co-Founder and Chief Executive Officer

Nairobi, KE & Burlington, VT

- Co-Founder of a Kenyan / US social enterprise that develops technology solutions to help organizations make smarter use of data to address important global challenges.
- Oversee growth of a global team of 70+ people that has delivered over \$15M USD in projects supporting global development partners.
- Contributed to the creation and development of OpenSRP, Reveal and Open Health Stack as digital global goods.
- Lead development of Ona Data and Akuko SAAS products that have been used to provide critical data collection and analysis services to 1000's of organizations globally including key UN organizations.
- Involved in the inception of the WHO SMART Guidelines including helping create in partnership with Google initial L4 reference applications.
- Developed Electronic Immunization Registry systems deployed in Zambia, Mauritania and Tunisia. Worked in partnership with GIZ and WHO to develop reference WHO SMART Guideline Applications for Immunizations.
- Developed national level community health worker applications (eCHIS) systems in Liberia, Uganda and Bangladesh that in aggregate are registered over 90M+ clients.
- Led design, development and implementation of digital health solutions in over a dozen countries.

The Earth Institute at Columbia University

July 2007-September 2013

ICT Director for the Modi Research Group & Millennium Villages Project

New York, NY

- Implement and provide strategic oversight of the Information & Communication Technology (ICT) sector for the Earth Institute.
- Led NYC and field based engineering team consisting of African regional and site based programming and technical staff
- Technical lead of the development of the Smart Registry platform for the NRHM in Karnataka, India.
- Led design and development of open source mobile data collection platform Formhub
- Design, develop and implement innovative mobile platforms including ChildCount+ - a mHealth platform that uses SMS text messages to monitor the health of women and children
- Development of a national scale SMS based clinic based monitoring system in Uganda
- Coordinate with field staff to design and implement ICT interventions in the Millennium Villages including the establishment of community radios, wifi-networks, school computer labs, medical record systems in clinics, ICT kiosks, phone charging booths and mobile phone application pilots

IESC Geekcorps Mali

October 2005-July 2007

Country Director

Bamako, Mali

- Chief of Party for a \$2.6 million dollar USAID Mali Communications for Development and Last Mile Initiative grant program
- Management of program office in French with 13 local staff and volunteer program with 10 international volunteers
- Implemented and developed appropriate technology solutions: LTSP Linux thin clients and multi-terminal computers, asynchronous Internet via USB key, low-power Geekcorps' Desert PC, video-over-Wi-Fi CanTV, bandwidth limiting RGBAN remote networking system, Wikipedia CD, Solar/LED lighting systems and low-cost solar FM Transmitter and studio

IESC Geekcorps & Digital Freedom Initiative in Senegal**May-August 2005***Geekcorps Volunteer*

Dakar, Senegal

- Developed/implemented a web/paper based accounting system for African merchants and women groups (*www.baolbaol.com*)

Inspidered, Inc – Web Development Firm**2003-2005***Co-founder*

Chicago, IL

- Provided web programming and design and database development services for non-profits in Chicago.

Meritum Corporation – Internet Startup**2002-2003***Chief Technology Officer*

Chicago, IL

- Oversaw the system design, programming, and billing support for a web-system built on open source standards that created a turnkey solution that allowed clients to sell self-replicating websites to over 20,000 paying members

Infocast Community – ISP & Internet Services Startup**2000-2002***Director of Creative Services*

Tucson, AZ

- Managed creative service team: 5 programmers, 2 designers and 2 support staff
- Oversaw the system design, programming, and billing support for a web-systems built on open source standards

EDUCATION**Thunderbird, School of International Management***MBA in International Management, GPA 3.80***Spring 2005**

- Ambassador Assistantship (1/3 tuition)
- Beta Gamma Sigma Business Honor Society
- Universidad Autónoma de Guadalajara Thunderbird Program (Guadalajara, Mexico, Summer 2004)

Knox College*Bachelor of Science Computer Science & Bachelor of Arts Integrated International Studies***Spring 2000**

- Lincoln Scholar (full tuition)
- Honors Dissertation: The State of Computer Education in Zimbabwe
- University of Zimbabwe: 8 month academic and cultural exchange (Harare, Zimbabwe, 1999)

AWARDS & HONORS

- 2018 Time Magazine 50 Genius Companies
- 2010 Time 100 List of the World's Most Influential People
- 2010 PopTech Social Innovation Fellow
- 2010 Knox College Young Alumni Award
- 2006 Tech Museum Award Laureate

Publications

- Garrett L Mehl, Martin G Seneviratne, Matt L Berg, Suhel Bidani, Rebecca L Distler, Marelize Gorgens, Karin E Kallander, Alain B Labrique, Mark S Landry, Carl Leitner, Peter B Lubell-Doughtie, Alvin D Marcelo, Yossi Matias, Jennifer Nelson, Von Nguyen, Jean Philbert Nsengimana, Maeghan Orton, Daniel R Otzoy Garcia, Daniel R Oyaole, Natschja Ratanaprayul, Susann Roth, Merrick P Schaefer, Dykki Settle, Jing Tang, Barakissa Tien-Wahser, Steven Wanyee, Fred Hersch. "A full-STAC remedy for global digital health transformation: open standards, technologies, architectures and content". Oxford Open Digital Health, Volume 1, 2023.
- Nandini Oomman, Garrett Mehl, Matt Berg, Rachel Silverman. "Modernising vital registration systems: why now?" The Lancet, Volume 381, Issue 9875, 2013, Pages 1336-1337.
- Berg, Matt, et al. "Cellular citizenship: invigorating development through mobile." Harvard International Review, vol. 34, no. 3, winter 2013, pp. 28+

- Caroline Asiimwe, David Gelvin, Evan Lee, Yanis Ben Amor, Ebony Quinto, Charles Katureebe, Lakshmi Sundaram, David Bell, and Matt Berg. “Use of an Innovative, Affordable, and Open-Source Short Message Service–Based Tool to Monitor Malaria in Remote Areas of Uganda”. The American Journal of Tropical Medicine and Hygiene. 2011 July 1; 85(1): 26–33.

Activities

- Thunderbird Global Council Member (2012-2014)
- WHO mHealth Technical Advisory Group (mTag) (2012-2014)

LANGUAGES

English (Native), French (Fluent), Spanish (Basic)

GEOGRAPHIC WORK EXPERIENCE

Afghanistan, Ethiopia, Ghana, India, Indonesia, Kenya, Mauritania, Malawi, Mali, Mexico, Nigeria, Rwanda, Senegal, Somalia, South Africa, Tanzania, Uganda, Rwanda, Sri Lanka, Viet Nam, Zambia, Zimbabwe

John Mwashuma

john.mwashuma@gmail.com | +254-716-647-845 | [LinkedIn](#) | [Github](#)

Summary

Senior Software Engineer with over 6 years of experience in full-stack development, cloud-native architectures, and microservices. Expert in leading backend teams, migrating applications to Kubernetes, and automating CI/CD pipelines. Proven track record of delivering high-impact solutions—such as the [TallyHo](#) election tallying system and the [Akuko BI platform](#)—through technical leadership, rigorous roadmap, sprint planning, and effective cross-functional team collaboration.

Professional Experience

Senior Software Engineer III / SRE Liason, Ona.io

Dec 2023 – Present

Key Achievements:

- **Team Leadership:** Lead backend microservices teams, and directed strategic roadmap planning and sprint sessions to prioritize feature development and ensure timely delivery.
- **Security & Scalability:** Integrated OpenID Connect with Keycloak and enhanced product reliability through proactive alerts using UptimeRobot, Prometheus, and OpsGenie.
- **Project Leadership – TallyHo:** As the technical lead for [TallyHo](#), an open source Django election tallying system, I coordinate with stakeholders—including the Libyan electoral body and UNDP—to deliver resilient, scalable, and timely election results, including the recent December 2024 Libyan Elections.
- **Cross-Functional Collaboration:** As an SRE liaison, I work closely with the SRE team to optimize infrastructure and ensure high system uptime. I have migrated microservices from EC2 to AWS EKS using Helm charts, [GoCD](#) and automated Docker image builds and deployments to AWS ECR via GitHub Actions, which improved performance and reduced costs. I also collaborate with the product and data engineering teams to drive continuous improvements on our [BI platform](#) data engineering workflows.

Key Projects:

- [Akuko](#): A BI platform for creating interactive maps, charts, and tables from diverse data sources.
- [TallyHo](#): An open-source Django election tallying system.

Tech Stack:

- Backend: Microservices (Golang, Python, Node.js/Typescript), Django/Python
- CI/CD: GithubActions/AWS ECR/GOCD
- Infrastructure: AWS EKS(K8's), Helm Charts, Terraform, Ansible, Packer
- Monitoring: Prometheus, Grafana, UptimeRobot, OpsGenie
- Databases: Clickhouse, PostgreSQL, [DuckDB](#), [Cube.js](#)
- Task Queues/Orchestrators: [Temporal](#), RabbitMQ, Celery, Redis

Key Achievements:

- **Research on Architecture Improvement:** Improved asynchronous task management by introducing [Temporal workflows](#) to replace legacy queuing systems.
- **End to End Ownership:** Built end-to-end features by developing React-based client interfaces and rolling out asynchronous backend tasks with [Temporal workflows](#) for long running tasks such as [PMTiles](#) generation for maps.
- **Developer Experience:** Refactored core microservices APIs from JavaScript to TypeScript to enhance maintainability and refactored IO bound services to Golang.
- **Data Engineering Collaboration:** Streamlined data engineering workflows by integrating the [Temporal orchestrator](#) for automated data ingestion and implementing a Python-based REST API semantic layer with [Cube.js](#), enabling rapid queries on [DBT-powered](#) data marts.
- **Training and Mentorship:** Conducted workshop training sessions on system functionality and code walkthroughs for the [TallyHo](#) project in preparation for elections.

Key Projects:

- [Temporal orchestrator](#): A durable execution orchestrator simplifies building scalable distributed systems.
- [Cube.js](#): An abstraction layer in modern data stack that sits between your data sources and data consumers.

Tech Stack:

- Frontend: React/Typescript
- Backend: Microservices (Golang, Python, Node.js/Typescript), Django/Python
- Mordern Data Stack Tools: [Cube.js](#), [DBT](#)
- Task Queues/Orchestrators: [Temporal](#), RabbitMQ, Celery, Redis

Key Achievements:

- **Architecture Performance Improvement:** Implemented compute and storage separation by deploying an internal ClickHouse database on Kubernetes to query Parquet files stored in any object store such as AWS S3, thereby eliminating the need for AWS Glue and Athena removing vendor lock-ins.
- **Product Stabilization and Reliability:** Improved the performance and reliability of our BI tool source creation workflows by implementing asynchronous CSV source creation using [DuckDB](#), which significantly boosted query performance.

Key Projects:

- [Akuko](#): A BI platform for creating interactive maps, charts, and tables from diverse data sources.

Tech Stack:

- Backend: Microservices (Golang, Python, Node.js/Typescript)
- Databases: Clickhouse, PostgreSQL, [DuckDB](#), [Cube.js](#), AWS Glue, AWS Athena

Software Engineer, Ona.io

Dec 2017 – Nov 2021

Key Achievements:

- Revamped CI/CD pipelines with GitHub Actions, reducing deployment times by 50%.
- Led a client project integrating [Onadata](#), [Enketo](#), [Apache Superset](#), and [Box](#) using a React/TypeScript frontend and Django REST backend.
- Established staging and production environments using Terraform, Ansible, and Packer.

Key Projects:

- [GRP](#): Resilience Platform, an online space to capture, access, co-create and advance the latest resilience knowledge.
- [Ona](#): A web based data collection app.
- [Onadata](#): An open source Django REST API for data collection.

Tech Stack:

- Frontend: React, Typescript, Clojure, ClojureScript, Reagent, Om
- Backend: Django Rest API (Python)
- CI/CD: Github Actions
- Infrastructure: AWS EC2, Terraform, Ansible, Packer
- Databases: PostgreSQL
- Task Queues/Orchestrators: RabbitMQ, Celery, Redis

Software Engineer Intern, Ona.io

Sep 2017 – Nov 2017

Key Achievements:

- Enhanced a ClojureScript-based frontend for a data collection tool by debugging and implementing feature improvements.

ASP.NET Developer (Contract), GrandLab Digital Fixers

Apr 2016 – May 2017

- Developed a shopping website by building both frontend and backend components.

IT Support Intern, Base Titanium Limited

Apr 2014 – Aug 2014

- Developed an internal web application using CodeIgniter to improve data management and reporting workflows.

Education

- **Bachelor of Science in Information Technology** 2012 – 2016
Technical University of Mombasa, Second Class Upper Division.
- **Master of Computational Intelligence - MCI, Artificial Intelligence** 2020 – Present
University of Nairobi (Coursework completed; project pending).

Faith Mutua

faithkanini18@gmail.com | +254726133241 | LinkedIn | Github

Professional Summary

Seasoned Quality Assurance Engineer with 7+ years of experience in delivering high-quality software. Proficient in various testing methodologies, including manual, automated, API, and performance testing. Demonstrated success in leading testing efforts, mentoring junior team members, and driving process improvements within collaborative, fast-paced environments. Proven expertise in analyzing complex requirements, designing comprehensive test plans and cases, and ensuring software quality within tight deadlines.

Skills

- Programming Languages: Java, Python, ClojureScript, JavaScript
- Testing Frameworks: Selenium, Espresso, Cypress, Appium, Cucumber, RestAssured
- Load Testing tools: JMeter, locust.io
- Continuous Integration/Continuous Delivery(CI/CD): Jenkins, Github Actions, Docker
- Software Development Methodologies: Agile(Kanban,Scrum), Waterfall
- Logs and Metrics Platforms: Sentry, Graylogs
- Computing Platforms: AWS, Azure
- Defect Tracking Systems: Github, Jira
- Test case management: Testrail
- Databases: Postgres and MySQL

Work Experience

Senior Software Engineer in Test I, Ona Systems Inc Jan 2023 – Present

- **Leveraged Espresso to automate mobile application tests**, boosting regression and end-to-end testing efficiency by 40% per release cycle.
- Championed the implementation of **Shift-Left testing** for **ona.io** using Clojure script decreasing debugging time by 30% and ensuring early defect detection and prevention.
- **Improved usability** of the Resilient Planet Initiative web application and chatbot by 45% through direct community **User Acceptance Testing**, driving key user experience and functionality enhancements.
- **Implemented a team upskilling initiative**, addressing identified skills gaps and leading to a 50% improvement in task efficiency through customized learning paths and a focused curriculum.
- **Mentored 3 junior team members** in best practices for Quality Engineering, enhancing their skills while boosting team efficiency by 25%.
- Employed proactive **error log analysis** with **Sentry** during technical support for ona.io, achieving an 85% reduction in client communication and significantly boosting customer satisfaction scores by 75%.

Software Engineer in Test, Ona Systems Inc Jan 2020 – Dec 2022

- Developed and maintained **automated test frameworks for database testing in Kubernetes-based CI/CD environments** using JUnit and Cucumber, achieving a 60% reduction in manual testing efforts and improved test coverage by 50%.
- Conducted **5+ performance tests** per month including stress, load and volume testing, using JMeter and locust, resulting in a 40% increase in concurrent user capacity and enhanced support for peak traffic.
- Performed thorough **mobile app performance testing**, improving app load times by 75% and identifying 10 critical performance bottlenecks, while also conducting thorough **device compatibility and bug root cause analysis** across 5 device configurations.
- **Investigated and reported** software defects using defect tracking systems (Github and Jira), ensuring accurate documentation and timely resolution of issues contributing to a 20% reduction in critical bugs.

- Ensured software quality by proactively **evaluating system requirements and design specifications**, identifying potential risks, and implementing targeted testing strategies to mitigate them, preventing costly defects and ensuring product stability.

Quality Assurance Engineer, Ona Systems Inc

Aug 2017 – Dec 2019

- Demonstrated expertise in **test case design** by constructing 500+ comprehensive test cases for complex software requirements, resulting in a 25% increase in test coverage and accelerated testing cycles for critical products.
- Executed **API automation testing** using Rest Assured, Java, and Cucumber to ensure robustness and reliability of backend systems, reducing manual testing time for APIs by 90% and increasing test coverage by 95%.
- Streamlined the **Quality Engineering process**, significantly reducing critical bugs by 80%, through thorough functionality, acceptance, regression, and compatibility testing.
- Actively participated in **architecture and design reviews** to prevent faulty design decisions before implementation.
- **Led process improvement** initiatives within **collaborative teams**, achieving a 15% reduction in testing cycle times and faster software delivery without compromising quality.

Education

Jomo Kenyatta University of Agriculture and Technology,
BSc in Mathematics and Computer Science

May 2009 – Dec 2012

Affiliations

Member of Ministry of Testing Community

Women Techmakers Ambassador Kenyan Chapter

COITE MANUEL

Founder & Chief Executive Officer | Crosscut

Washington, DC | coite@crosscut.io | +1-703-727-8784 | crosscut.io

1. Executive Summary

Coite Manuel is the Founder and Chief Executive Officer of Crosscut, a certified Digital Public Good and DHIS2 Tier 2 Strategic Technology Partner that powers the WHO AFRO ESPEN Geospatial Microplanner and supports geospatial campaign planning across 43 countries. He brings more than 20 years of experience spanning global health supply chain strategy, digital health system design, interoperable platform development, and multi-country implementation leadership — including five years as Senior Supply Chain Advisor in USAID's Global Health Office.

He brings direct operational experience integrating campaign data systems across NTD mass drug administration, immunization, malaria IRS/ITN, and vitamin A supplementation programs, and has coordinated complex, multi-country digital health initiatives with Ministries of Health, UNICEF, WHO/ESPEN, Gavi, and implementing partners. Crosscut has led the development of open-source, DHIS2-compatible production integrations with ODK, CommCare, and the WHO ESPEN Portal. The company holds formal DPG status and a signed DHIS2 Technology Partner Agreement, directly supporting UNICEF's ICR objectives of interoperability, sustainability, and government institutionalisation.

2. Core Technical Competencies

- Multi-country program management
- Geospatial microplanning & GIS analytics
- NTD MDA, immunization & malaria campaign systems
- Digital Public Goods design & stewardship
- Stakeholder engagement: MoH, UN, Gavi, USAID
- Capacity building & knowledge transfer
- Campaign data integration & interoperability
- DHIS2 production integration (Tier 2 Partner)
- ODK & CommCare data connectors (production)
- WHO ESPEN / JAP reporting workflows
- Open-source platform architecture & licensing
- Low-connectivity & offline-first deployment

3. Professional Experience

Crosscut | Founder & Chief Executive Officer | Washington, DC | Apr 2017 – Present

Founded and leads a digital health technology company specialising in geospatial campaign planning, population estimation, and interoperable data integration for public health programs in low- and middle-income countries (LMICs). Manages strategy, product direction, client delivery, and partner engagement across concurrent multi-country programmes funded by UNICEF, WHO, Gavi, USAID/PMI, and the Gates Foundation.

- Directs a multidisciplinary team of engineers, data scientists, and global health professionals operating across 10+ Sub-Saharan African countries simultaneously, under a disciplined agile development model (quarterly planning, two-week sprints, daily standups, formal code review).
- Leads all client-facing engagement with Ministries of Health, UN agencies, and international donors — serving as principal contact for UNICEF HQ, WHO AFRO/ESPEN, Gavi, and USAID/PMI — coordinating technical delivery across stakeholder groups in both English and French-speaking contexts.
- Oversees design and maintenance of open-source production integrations with DHIS2, ODK, CommCare, and the WHO ESPEN Portal, enabling campaign data to flow from community-level collection tools into national health management information systems.
- Positioned Crosscut as a DHIS2 Tier 2 Strategic Technology Partner (signed agreement with HISP Centre, University of Oslo) and as a verified Digital Public Good (DPG ID: GID0090906, Digital Public Goods Alliance), directly aligning with UNICEF's ICR interoperability and sustainability requirements.
- Secured recognition as 2023–2024 COR-NTD Innovator of the Year (Task Force for Global Health) and selection as a Gavi 6.0 global geospatial partner (with VillageReach), underscoring domain credibility across NTD and immunization programs.

U.S. Agency for International Development (USAID) | Senior Supply Chain Advisor, Global Health Office | Washington, DC | Jan 2012 – Mar 2017

- Managed a broad portfolio of projects related to the strategy, policy, and operational improvement of USAID's \$1B per year global health pharmaceutical and medical supplies procurement and logistics programme spanning multiple countries and health commodity areas.
- Provided senior technical guidance on supply chain design, analytics platforms, and operational strengthening in collaboration with implementing partners, host-government Ministries of Health, and multilateral counterparts.
- Contributed subject-matter expertise at the intersection of supply chain management, health system strengthening, and programme data use — experience directly applicable to embedding campaign data workflows within national reporting cycles.

Independent Supply Chain Consultant | Washington, DC | May–Dec 2011

- Provided technical expertise and strategic advisement on the distribution of medical supplies in developing countries for the World Bank and John Snow Inc. (JSI).

Accenture | Manager, Supply Chain Strategy Practice | Jan 2004 – Sep 2006 | Ranked Exceptional, Top 5%

- Led supply chain strategy engagements across the public and private sectors, achieving a top-5% internal performance ranking. Selected project highlights:
- Department of Health and Human Services (HHS): Led study to determine optimal inventory storage and distribution strategy; developed warehouse consolidation plan representing a 22% reduction in annual operating expense.
- Office supply retailer: Performed inventory optimisation analysis reducing stock levels by more than 20% while maintaining service levels.
- Fashion retailer: Led distribution capacity planning and analysis for domestic processing centres, modelling processing capabilities against forecasted demand.
- Clothing retailer: Performed distribution network optimisation to support a doubling of store count over five years.
- Electronics retailer: Determined optimal product flow path from vendor to customer through mathematical optimisation.

Accenture | Analyst, Supply Chain Strategy Practice | Jan 2001 – Sep 2001

- Contributed to supply chain strategy engagements as an analyst, building foundational expertise in supply chain modelling, network design, and operational analytics.

4. Selected Project Experience (ICR-Relevant)

WHO AFRO ESPEN Geospatial Microplanner — Lead Developer & Maintainer

Partners WHO AFRO / ESPEN, Gates Foundation

Geography 43 Sub-Saharan African countries, all WHO AFRO member states

Relevance ICR Phase 5 directly requires two-way integration with this same tool

- Built and maintains the WHO ESPEN Geospatial Microplanner (espen.afro.who.int) — the primary sub-district NTD MDA planning tool across 43 countries — as a white-label deployment of the Crosscut platform. This is the exact tool the ICR must establish two-way integration with under Phase 5.
- Developed the ESPEN Schisto Mapper Tool (piloted Senegal) — translates schistosomiasis survey and environmental data into actionable sub-district treatment maps within the ESPEN Portal.
- Participated in WHO workshop (Brazzaville, December 2025) with representatives from 13 countries to demonstrate the Microplanner and gather country feedback, providing ground-level insight into NTD microplanning workflow gaps across the region.

DHIS2 Microplanning App & Tier 2 Technology Partnership

Partners HISP Centre, University of Oslo; 30+ Ministry of Health DHIS2 administrators

Geography Multi-country; 30+ MoH users in Sub-Saharan Africa

Relevance ICR Phase 5 DHIS2 connector; ICR institutionalisation within national HMIS

- Holds a signed Technology Partner Agreement with the HISP Centre designating Crosscut as a DHIS2 Tier 2 Strategic Technology Partner (Digital Public Good) — covering API-level integration, joint reference architectures, coordinated country deployments, and joint donor/government engagement.

- Built and maintains a dedicated DHIS2 Microplanning app on the DHIS2 App Hub enabling national administrators to create sub-district catchment areas directly within DHIS2 without GIS expertise; DHIS2 v41 (May 2024) ships with native support for Crosscut catchment visualisation.
- Trained 50+ Ministry of Health staff across multiple countries at the 2023 DHIS2 Academy to independently generate catchment areas and microplans within their national DHIS2 instances.

NTD Mass Drug Administration Campaign Systems — Multi-Country

Partners	Sightsavers, The Carter Center, WHO ESPEN, Task Force for Global Health
Geography	Nigeria, Liberia, Guinea-Bissau, Haiti, plus 2026 expansion to 8 additional countries
Relevance	ICR core use case: NTD MDA campaign data integration, coverage tracking, JAP reporting
	• Nigeria (Sightsavers / Kogi State MoH): Led schistosomiasis MDA microplanning across Lokoja, Kogi, and Ibaji LGAs, targeting 152,724 children aged 5–14. Delivered accessibility analysis, supervisory area delineation, and a CommCare integration for real-time supervision monitoring. • Liberia (Sightsavers / MoH): Piloted ESPEN Geospatial Microplanner across three counties (Maryland, Grand Cape Mount, Montserrado), building catchment-based microplans for NTD MDA campaigns from geocoded facility data. • Haiti (The Carter Center / MoH): Supported lymphatic filariasis MDA campaigns; Crosscut population estimates directly informed a shift from fixed-point to door-to-door drug distribution, demonstrating data-to-decision impact on campaign design. • Nigeria (The Carter Center): Production ODK integration for MDA supervision — field supervisors log visits via ODK; Crosscut ingests and maps data geographically, enabling program managers to track field team coverage in real time. • Liberia, Guinea-Bissau, Nigeria (Sightsavers): Production CommCare integration for MDA supervision monitoring — field supervisors log via CommCare; Crosscut surfaces supervision coverage alongside catchment boundaries.

Immunization Campaign Microplanning & Supply Chain Analytics

Partners	UNICEF, Gavi, VillageReach
Geography	The Gambia, Ethiopia; Gavi global partner framework (20+ countries)
Relevance	ICR immunization campaign data flows, Gavi/COVAX reporting, SIA microplanning
	• The Gambia (UNICEF / MoH EPI): Supported digitised electronic microplanning for immunization campaigns, helping government make evidence-based platform decisions. Platform supports SIA planning, vaccination post identification, session planning, and population-benchmarked coverage estimation. • Ethiopia (UNICEF, Gavi): Selected as a Gavi global partner in supply chain analytics (2022, prequalified Ethiopia). Integrated with UNICEF's Stock Management Tool (SMT), in active use in 20+ countries, enabling batch-level vaccine visibility from health facility to central medical store — supporting expiry prevention and stockout detection. • Selected as Gavi 6.0 Technical Assistance global partner in geospatial microplanning (with prime partner VillageReach, 2024–2025).

Malaria IRS & ITN Campaign Planning — USAID/PMI Evolve

Partners	Abt Associates (prime), USAID/PMI
Geography	21-country programme; design sessions in Uganda and Ghana in Year 1
Relevance	Campaign planning for large-scale public health interventions; IRS/ITN data flows
	• Selected as resource partner on USAID's PMI Evolve Project (Abt Associates prime), a five-year IRS program spanning 21 countries. Conducted design sessions and user testing in Uganda and Ghana for IRS supply estimation and ITN distribution planning. • Developed ML-based building classification (>85% accuracy, identifying spray-eligible structures) generating printable PDF operational maps for IRS campaigns — demonstrating AI-assisted campaign data use in LMIC field contexts.

5. Digital Public Goods & Open Source Contributions

- Verified Digital Public Good — Crosscut is registered with the Digital Public Goods Alliance (DPG ID: GID0090906; registry: digitalpublicgoods.net/r/crosscut). All 11 DPG Standard indicators met, including open licensing (CC0 1.0), platform independence, technical documentation, non-PII data extraction pathways, privacy

compliance (GDPR), and protection from harassment. Output datasets also published on the Humanitarian Data Exchange (HDX).

- DHIS2 App Hub — Maintains the dhis2-crosscut-app (github.com/crosscutio), enabling national DHIS2 administrators to import Crosscut catchment areas directly into their DHIS2 instances. More than 30 Ministry of Health users have incorporated Crosscut data into DHIS2 instances across Sub-Saharan Africa.
- Field-Kit (released March 2026) — Open-source geospatial data preparation toolkit published under Apache 2.0 license, available on GitHub.
- 6 additional public repositories on GitHub (github.com/crosscutio) covering platform integrations and data pipelines.
- Organisational users of Crosscut's open datasets include the Ministry of Health EPI programme in The Gambia, the World Health Organization, and 30+ MoH users who have incorporated the data into DHIS2 and other national health information systems.

6. Country & Stakeholder Engagement

Ministries of Health — Direct Engagement (10+ Countries)

Country	MoH Counterpart	Programme Area
Nigeria	Kogi State Ministry of Health	Schistosomiasis MDA microplanning
Ethiopia	Ministry of Health (via UNICEF)	Immunization supply chain design
Uganda	Ministry of Health (via PMI Evolve)	Malaria IRS planning
Ghana	Ministry of Health (direct + PMI Evolve)	Malaria IRS; CHPS zone mapping
Liberia	Ministry of Health	ESPEN Microplanner pilot; NTD MDA
Guinea-Bissau	Ministry of Health	App training, deployment, MDA support
Burundi	Ministry of Health	Onchocerciasis endemicity mapping
The Gambia	Ministry of Health, EPI	Supply chain design; microplanning
Haiti	Ministry of Health (via Carter Center)	Lymphatic filariasis MDA
DRC	Ministry of Health (via Unlimit Health)	Schistosomiasis boundary delineation

UN Agencies, Donors & Multilateral Partners

- UNICEF — Active engagement at HQ and Country Office levels on digital health, geospatial microplanning, immunization supply chain, and NTD integration programmes.
- WHO / ESPEN — Lead developer and partner for the ESPEN Geospatial Microplanner (43 countries); participant in WHO technical workshops on NTD microplanning.
- Gavi — Prequalified global partner in supply chain analytics (2022) and geospatial microplanning (6.0 framework, 2024–2025) in collaboration with VillageReach.
- USAID / PMI — Resource partner on PMI Evolve Project (Abt Associates prime), a five-year IRS programme across 21 countries.
- Gates Foundation — Grand Challenge grant recipient (selected from 1,100+ applications, 2020); ongoing technical collaboration on NTD microplanning.
- Task Force for Global Health — COR-NTD Innovator of the Year (2023–2024); technical partner for NTD campaign planning.

7. Education & Professional Recognition

M.S., Industrial Engineering | Georgia Institute of Technology | 2000

B.S., Applied Mathematics & Mathematical Economics | Hampden-Sydney College | 1999

Selected Recognition

2025 Gavi 6.0 global geospatial microplanning partner (with VillageReach)

2024	Certified Digital Public Good — Digital Public Goods Alliance (DPG ID: GID0090906)
2024	Selected as 1 of 10 technology companies to present at Alliance for Malaria Prevention annual meeting
2023	COR-NTD Innovator of the Year — Task Force for Global Health (retained 2023–2024)
2022	DHIS2 App of the Year finalist — DHIS2 Annual Conference
2022	Gavi global partner in supply chain analytics (prequalified Ethiopia)
2020	Gates Foundation Grand Challenge grant — selected from 1,100+ applications

JAMES McKINNON, M.A.

Supply Chain Data Analyst & Geospatial Health Specialist | Crosscut

St Paul's Bay, Malta | james@crosscut.io | +1 260-433-6426

1. Executive Summary

Global health supply chain analyst and geospatial data specialist with 10+ years of experience delivering analytics and field operations across 10+ countries in Sub-Saharan Africa, South Asia, and beyond. At Crosscut, James leads data-intensive analytical work including GIS-based operational map production for health campaigns, raster analysis, and network modelling for vaccine supply chains. He has led UNICEF-funded supply chain assessments in Ethiopia and The Gambia — including personal geocoding of The Gambia's entire national vaccination network — managed emergency health campaign logistics for Médecins Sans Frontières in South Sudan (including a measles vaccination cold chain strategy), and coordinated supply chain data across 38 countries as part of the USAID Global Health Fellows Program.

2. Core Technical Competencies

- Supply chain analytics & quantitative modelling
- Immunization supply chain assessment
- UNICEF programme support (field & HQ)
- Field data collection & logistics management
- Multi-country data coordination (38 countries)
- GIS / geospatial analysis (QGIS, ArcMap, Python)
- Raster & network analysis for health campaigns
- Dashboard development (Tableau, Power BI, MicroStrategy)
- Mixed-methods research design & implementation
- Emergency health campaign logistics

3. Professional Experience

Crosscut | Supply Chain Data Analyst | The Gambia; Senegal; Ethiopia | 2018–Present (intermittent)

- Conducts advanced GIS data analysis including aggregation of raster-based datasets summarising environmental suitability scores for disease-causing parasites, network analyses assessing optimal vaccine supply chain design, and automated production of operational maps supporting health campaigns — using Python (geopandas, rasterio, osgeo) and QGIS.
- Led data collection and analysis for a UNICEF-funded assessment of the Ethiopian government's public health supply chain, using a mixed-methods approach combining quantitative supply chain modelling with site visits to all government medical stores (assessing accessibility, warehouse design, and inventory management policies).
- Personally geocoded all sites within The Gambia's public vaccination network: visited each facility to collect GPS coordinates, photographs, and cold chain equipment inventories within a tightly managed schedule.
- Collected and analysed data for a UNICEF-funded country-wide immunization supply chain assessment in The Gambia, incorporating quantitative supply chain modelling and qualitative inputs from UNICEF and Ministry of Health personnel, including scenario analysis of alternative network designs.

Médecins Sans Frontières (MSF) | Project Supply Chain Manager | Upper Nile State, South Sudan | Jan–Oct 2022

- Served as Head of Supply for the Maban project (outpatient clinic and 7 outreach sites), managing a team of 6 staff with responsibility for procurement, inventory management, forecasting, and cargo flight coordination.
- Developed and executed a cold chain strategy for an emergency measles vaccination campaign in a remote area more than half a day from any active cold chain equipment — using passive cold chain devices for the full 6-day campaign duration, with detailed ice pack management and replenishment planning.
- Coordinated delivery and storage of 90+ tonnes of non-food item material for a flood emergency distribution, requiring cargo flight coordination with the MSF Juba office and synchronisation with field distribution teams.

HOPE Foundation for Women and Children of Bangladesh | Program Manager | Cox's Bazar, Bangladesh | Mar–Sep 2019

- Led programme design and management of the Emergency Response Team project, keeping mobile medical teams on standby for Rohingya refugee and host Bangladeshi communities during Cyclone Fani and subsequent monsoon flooding.
- Represented HOPE at the UN-coordinated Logistics Sector and Health Sector Emergency Preparedness Working Group, coordinating with UN agencies and partner organisations.

USAID Global Health Fellows Program II | Supply Chain Data Analyst | Washington, DC | Mar 2015–Dec 2017

- Coordinated collection of supply chain data for health commodities across 38 countries under the SCMS and DELIVER projects, enabling visibility into in-country supply chains and building a model for future implementing partner data submissions.
- Served as primary point of contact for supply chain technical assistance and pharmaceutical shipments to Benin and Mali, including stock monitoring, importation support, and USAID budget cycle coordination.
- Developed MicroStrategy dashboards consolidating logistics data from multiple sources to support the Commodities Security and Logistics Division; contributed to M&E procedures for USAID Global Health Supply Chain awards.

JVA Consulting LLC | Research & Evaluation Assistant | Denver, CO | Nov 2013–Feb 2015

- Managed 3 projects from start to completion as Project Lead, including a county-wide service provider assessment. Developed Social Network Analysis, Excel-based dashboards, and trained staff on data analytics and M&E practices.

4. Selected Project Experience (ICR-Relevant)

UNICEF Ethiopia Public Health Supply Chain Assessment

Partners	UNICEF, Ethiopian Federal Ministry of Health
Scope	Site visits to all government medical stores; national supply chain assessment
Methods	Mixed methods: quantitative supply chain modelling + qualitative field assessment
Relevance	ICR Phases 3 & 4: MoH data flows, national reporting integration, country engagement
• Applied mixed-methods approach combining quantitative supply chain modelling with direct site visits to all government medical stores — generating facility-level data on accessibility, warehouse design, inventory management, and cold chain performance directly analogous to the ICR's Location and Campaign Metadata data model elements.	

UNICEF Gambia Immunization Supply Chain Assessment & Vaccination Network Geocoding

Partners	UNICEF, Gambia Ministry of Health EPI Programme
Scope	Full national vaccination network; country-wide supply chain assessment
Methods	Personal site visits; GPS coordinates; cold chain inventory; quantitative + qualitative analysis
Relevance	ICR data model: Location resource, GPS facility data, immunization supply chain reporting
• Personally visited every site in The Gambia's national vaccination network to collect GPS coordinates, photographs, and cold chain equipment inventories — producing a geocoded facility dataset directly analogous to the ICR Location resource requirements for health facility catchment area and service delivery point data. • Conducted a country-wide immunization supply chain assessment incorporating UNICEF and MoH inputs, quantitative modelling, and scenario analysis of alternative network designs, informing national EPI programme planning.	

Emergency Measles Vaccination Campaign — Cold Chain Strategy (MSF, South Sudan)

Partners	Médecins Sans Frontières, Upper Nile State health authorities
Scope	6-day vaccination campaign, remote area; passive cold chain management throughout
Relevance	ICR campaign execution understanding: Services Received, cold chain metadata, campaign logistics
• Designed and executed a cold chain strategy enabling a measles vaccination campaign in a setting without active cold chain access, managing passive cold chain devices and ice pack replenishment for 6 days — providing hands-on operational understanding of the campaign execution data flows the ICR is designed to capture and standardise.	

5. Open Source & Data Contributions

- Produced automated operational maps for public health campaigns distributed to implementing partners and health authorities across Sub-Saharan Africa through Crosscut's platform.
- Contributed to supply chain data infrastructure under USAID's SCMS and DELIVER projects, supporting data visibility and reporting standardisation for 38 countries.

6. Country & Stakeholder Engagement

Countries with Direct Programme Engagement

Country	Organisation / Context	Programme Area
The Gambia	UNICEF / Ministry of Health EPI	Immunization supply chain; vaccination network geocoding
Ethiopia	UNICEF / Federal Ministry of Health	Public health supply chain assessment
South Sudan	Médecins Sans Frontières	Measles vaccination campaign; emergency logistics
Bangladesh	HOPE Foundation / UN Agencies	Emergency health response; cyclone preparedness
Benin & Mali	USAID / SCMS / DELIVER	Supply chain technical assistance; stock monitoring

Additional country experience (data analysis, site-based work): Ghana, Tanzania, Senegal (Crosscut platform analytics); India (evaluation internship, Odisha); Russia (academic abroad).

Key Partner Engagements

- UNICEF — direct project delivery in Ethiopia and The Gambia on supply chain assessments.
- Médecins Sans Frontières — project supply chain management and emergency vaccination campaign, South Sudan.
- USAID / Global Health Fellows Program II — 38-country supply chain data coordination; primary liaison for Benin and Mali.
- UN Logistics Sector & Health Sector Emergency Preparedness Working Group — represented HOPE Foundation, Bangladesh.

7. Education & Technical Skills

M.A., International Development | Josef Korbel School of International Studies, University of Denver | 2014

- Relevant coursework: Information Management in Humanitarian Crises; Field Operations for Humanitarian Assistance; Mobile Technology for International Development; Introduction to Epidemiology; Time Series Analysis; Econometrics for Decision Making.

B.A., Mathematics and Philosophy | Wabash College | 2007 | Summa cum laude, Phi Beta Kappa

Languages: English (native speaker)

Technical tools: Python (geopandas, rasterio, osgeo), R, Stata, SQL, QGIS, ArcMap, Tableau, Power BI, MicroStrategy, Excel

CLARA R. BURGERT, PhD, MPH

Infectious Disease Epidemiologist & Geospatial Health Specialist | Crosscut

Silver Spring, MD | crburgert@gmail.com | +1 202-203-8236 | English & French (fluent)

1. Executive Summary

PhD-trained infectious disease epidemiologist with 20+ years of experience applying advanced quantitative and spatial methods to neglected tropical disease programs, immunization campaigns, malaria surveillance, and public health intervention evaluation across 20+ countries. Clara currently leads multi-country evaluations of geospatial microplanning for community drug distribution programs and serves as project lead for Francophone country activities, managing stakeholder engagement with Ministries of Health and implementing partners in French. She brings 100+ multi-country epidemiological studies, 30+ peer-reviewed publications, and direct technical engagement with WHO, Gavi, USAID, and Ministries of Health to the ICR assignment. Fluent in English and French, she is positioned to lead all French-language country engagement and capacity-building activities required by the ToR.

2. Core Technical Competencies

- NTD programme epidemiology (20+ countries)
- Geospatial & geostatistical modelling (R)
- Infectious disease surveillance & evaluation
- Data pipelines, QA & reproducible workflows
- Study design, protocols & statistical analysis
- French-language MoH & partner engagement (fluent)
- Model-based geostatistical models (MBG)
- WHO & Ministry of Health technical engagement
- Capacity building (10+ regional trainings)
- Evidence-to-policy translation for WHO/Gavi/USAID

3. Professional Experience

Epidemiology Consultant (Self-employed) | Remote | Oct 2025–Present

- Leads executive-level, multi-country evaluations of geospatial microplanning initiatives for community drug distribution programs; designs evaluation frameworks, data collection tools, and integrates quantitative and qualitative data to assess implementation effectiveness, coverage implications, and scalability.
- Serves as project lead for Francophone country activities, managing timelines, deliverables, and stakeholder engagement with Ministries of Health and implementing partners in French.
- Supports digital health tool adoption through stakeholder interviews, usability assessments, and training development; translates user feedback into actionable recommendations.

RTI International | Epidemiologist, Manager | Washington DC / Remote | Oct 2017–May 2025

- Designed, implemented, and analysed infectious disease surveillance and epidemiologic evaluation studies across 20+ countries, supporting neglected tropical disease programmes in collaboration with Ministries of Health, WHO, and implementing partners.
- Served as technical and project lead for 100+ multi-country studies: authored study protocols, statistical analysis plans, and data collection tools; coordinated multidisciplinary teams and managed timelines, deliverables, and partner communications.
- Led advanced statistical, spatial, and model-based geostatistical analyses in R to assess infectious disease risk and evaluate public health intervention performance; developed, calibrated, and validated MBG models against observed epidemiological data, informing national surveillance strategies and WHO policy guidance.
- Developed and maintained reproducible data pipelines and quality assurance scripts in R supporting data cleaning, integrity monitoring, and analysis across large multi-country datasets.
- Produced technical reports, policy briefs, and analytical summaries for donors, Ministries of Health, and global health partners; supported 20+ conference presentations and 30+ peer-reviewed publications.
- Supervised 3 staff; provided technical training in applied epidemiology, surveillance analytics, and data visualisation.

ICF International | Technical Specialist | Rockville, MD | Jan 2011–Oct 2017

- Contributed to applied epidemiology, surveillance, and evaluation activities across HIV, malaria, vaccine-preventable diseases, and MCH programmes under USAID-funded initiatives including the Demographic and Health Surveys (DHS) Programme and MEASURE Evaluation.
- Conducted statistical and spatial analyses assessing disease burden, surveillance performance, and programme outcomes; produced evidence-based recommendations for WHO, Gavi, and USAID stakeholders.
- Collaborated with Ministries of Health, academic partners, and multidisciplinary teams across multiple countries to translate epidemiological findings into programme planning and policy decisions.
- Facilitated 10+ regional trainings in epidemiology, GIS, and data visualisation, strengthening analytical capacity among Ministry of Health and partner staff.
- Managed technical deliverables, timelines, and budgets for large, multi-year USAID-funded programmes; supervised 3 staff; served as primary technical contact for partner and donor engagements.

Blue Raster | Research Analyst | Calverton, MD | 2009–2010

- Conducted spatial and epidemiologic analyses for malaria surveillance and prevalence under USAID-funded programmes.
- Managed field data collection in Haiti with UNICEF, supporting emergency response surveillance.

Catholic Relief Services (CRS) | International Development Fellow | Dakar, Senegal | 2008–2009

- Designed and implemented a community-based health survey covering HIV, malaria, and nutrition in collaboration with the Senegal Ministry of Health; supported indicator-based surveillance, analysis, and reporting for donors and national stakeholders.

Centers for Disease Control (CDC) | Research Assistant | Atlanta, GA & Kisumu, Kenya | 2007–2008

- Supported infectious disease surveillance studies through data analysis, reporting, and interpretation in Atlanta and in the field in Kisumu, Kenya.

4. Selected Project Experience (ICR-Relevant)

Geospatial Microplanning Evaluation — Community Drug Distribution Programs (Multi-Country)

Partners	Ministries of Health; implementing partners (current engagement)
Scope	Multi-country evaluation: geospatial microplanning for NTD community drug distribution
Language	Francophone country activities led entirely in French
Relevance	ICR Phase 3: capacity building, Francophone country engagement, implementation evaluation
• Leads executive-level evaluations of geospatial microplanning for NTD community drug distribution programs — directly relevant to the ICR's requirement for evaluating and institutionalising campaign data workflows in pilot countries. • Serves as Francophone country project lead — managing MoH and implementing partner engagement in French — meeting the ToR's explicit requirement for French-language capacity-building workshop leadership.	

NTD Programme Surveillance & Epidemiological Studies — RTI International, 20+ Countries

Partners	WHO, Ministries of Health, USAID, implementing partners
Scale	100+ multi-country studies; 20+ countries; 30+ peer-reviewed publications
Methods	Advanced statistical, spatial, and geostatistical modelling in R; MBG model development
Relevance	ICR NTD domain expertise; campaign coverage evaluation; data standards and quality assurance
• Led 100+ multi-country NTD epidemiological studies across lymphatic filariasis, schistosomiasis, trachoma, onchocerciasis, and soil-transmitted helminths — providing deep understanding of the NTD disease landscape, treatment protocols, and data standards the ICR must accommodate and standardise. • Developed, calibrated, and validated model-based geostatistical models against population-level epidemiological data across 30+ countries, informing national surveillance strategies and WHO policy guidance — directly relevant to the ICR's analytics layer and data quality objectives.	

PhD Research: Trachoma Geostatistical Modelling — London School of Hygiene & Tropical Medicine

Degree	PhD, Infectious and Tropical Diseases Epidemiology (awarded Sept 2025)
Thesis	Exploratory geospatial analysis and modelling to support trachoma elimination and surveillance

Scope	Model-based geostatistical models across 30+ trachoma-endemic countries
Relevance	ICR: geospatial NTD disease modelling; evidence-to-WHO-policy translation
	Developed and calibrated model-based geostatistical models to population-level trachoma prevalence data across 30+ endemic countries — the same geostatistical modelling approach the ICR analytics layer will use to translate campaign coverage data into actionable microplanning intelligence for Ministries of Health.
DHS Programme & MEASURE Evaluation — ICF International, USAID (Multi-Country)	
Partners	USAID, Ministries of Health, WHO, Gavi
Scope	Multi-country HIV, malaria, VPD, MCH surveillance; 10+ regional trainings delivered
Relevance	ICR: health information systems, national data flows, capacity building, MoH engagement
	- Contributed to the DHS Programme and MEASURE Evaluation — building a multi-decade understanding of national health information systems, data flows, and reporting standards that underpin ICR design requirements. - Facilitated 10+ regional trainings in epidemiology, GIS, and data visualisation for Ministry of Health and partner staff across multiple countries — directly applicable to the ICR Phase 3 capacity-building mandate.

5. Selected Peer-Reviewed Publications

- Harte A, Sasanami M, Burgert-Brucker CR, et al. (2025). Using Model-Based Geostatistics to Refine Population-Based Estimates of Trachoma Prevalence. *American Journal of Tropical Medicine and Hygiene*.
- Sasanami M, Burgert-Brucker CR, et al. (2025). Understanding the impact of covariates for trachoma prevalence prediction using geostatistical methods. *BMC Global and Public Health*.
- Burgert-Brucker CR, Adams MW, Solomon AW, Harding-Esch EM. (2022). Community-level trachoma ecological associations and geospatial analysis: A systematic review. *PLoS NTD*.
- Burgert-Brucker CR, et al. (2020). Risk factors associated with failing pre-transmission assessment surveys in lymphatic filariasis elimination programs. *PLoS NTD*.
- Burgert CR, Bradley SEK, Arnold F, Eckert E. (2014). Improving estimates of insecticide-treated mosquito net coverage using geographic coordinates. *Malaria Journal*.

Full list (30+ publications): orcid.org/0000-0002-6001-4960

6. Country & Stakeholder Engagement

Select Countries with Direct Programme Engagement

Country	Organisation / Context	Programme Area
Senegal	Catholic Relief Services / Ministry of Health	Community health survey; HIV, malaria, nutrition
Haiti	Blue Raster / UNICEF	Emergency response surveillance; field data collection
Kenya	Centers for Disease Control (CDC)	Infectious disease surveillance, Kisumu
20+ LMICs	RTI / ICF / USAID	NTD surveillance, malaria, DHS, MEASURE Evaluation

Key Institutional Partnerships

- WHO — Direct technical engagement on NTD surveillance and policy guidance through RTI International; trachoma PhD research informing WHO elimination targets.
- UNICEF — Field data collection in Haiti for emergency response surveillance (Blue Raster); ongoing Crosscut engagement.
- Gavi — Evidence production for vaccine-preventable disease programme planning (ICF / USAID collaboration).
- USAID — Technical lead on DHS Programme and MEASURE Evaluation; 10+ year partnership across multiple country programmes.
- Ministries of Health — Direct engagement in 20+ countries across NTD, immunization, malaria, and MCH programmes; Francophone country lead for current microplanning evaluations.

7. Education

PhD, Infectious and Tropical Diseases Epidemiology | London School of Hygiene & Tropical Medicine | 2025

- Thesis: 'Exploratory geospatial analysis and modelling to support trachoma elimination and surveillance.'
Developed and calibrated MBG models across 30+ endemic countries.

M.P.H., Global Epidemiology | Rollins School of Public Health, Emory University | 2008

B.S., International Relations (Minor: Biology) | St. Catherine University | 2003

Languages: English (fluent) | French (fluent)

Technical Skills: R, Stata, ArcGIS, QGIS, ODK, MS Excel, Git; statistical and geostatistical modelling; data pipeline development; protocol and SAP design

SAMUEL HOOGEWIND, B.S.

Software Engineer | Crosscut
Washington, DC | samuel.hoogewind@gmail.com | (616) 406-9208

1. Executive Summary

Software engineer with expertise in cloud-based geospatial data pipelines, population modelling algorithms, and frontend visualisation for global health applications. At Crosscut, Sam engineers scalable AWS data pipelines supporting real-time geospatial data access for public health planning across 45+ African countries, maintaining 99% + system uptime. He designed the foundational population raster algorithm now serving as the core data layer for downstream infrastructure analytics across the platform, and builds QGIS plugins to automate internal geospatial workflows. His work directly supports the data infrastructure that feeds into campaign microplanning and the analytics layer the ICR will integrate with.

2. Core Technical Competencies

- AWS data pipeline engineering (Python, R)
- Geospatial data processing (45+ African countries)
- Population raster algorithm design
- React / JavaScript frontend development
- QGIS plugin development (Python automation)
- ECS / Docker cloud infrastructure
- High-availability backend systems (99%+ uptime)
- Data integrity across multi-region datasets

3. Professional Experience

Crosscut | Software Engineer | Washington, DC | June 2025–Present

- Engineers scalable data pipelines on AWS using Python and R to process and manage geospatial datasets for 45+ African countries, supporting real-time data access for hundreds of users.
- Designed a foundational population raster algorithm, migrating legacy hexagonal models to a high-performance grid-based architecture that now serves as the core population data layer for downstream infrastructure analytics across the platform.
- Maintains and updates the multi-component React/JavaScript frontend, optimising the visualisation of large-scale geographic datasets and improving interface responsiveness for complex user queries.
- Builds custom Python plugins for QGIS to automate internal geospatial workflows, reducing manual data processing time and increasing delivery speed.
- Optimises backend data structures for high-intensity geographic processing, maintaining 99%+ system uptime and ensuring data integrity across disparate regional datasets.

Salesforce Analyst, Calvin University | Remote | Dec 2024–Jun 2025

- Automated data extraction and integration for 15,000+ alumni records using Apex and Python; built custom Apsona reports and scripts reducing manual research time by 200%.

Development and Advancement Intern, Calvin University | Grand Rapids, MI | May 2023–Jun 2024

- Engineered data structures for a 10,000-record database migration; designed custom object mappings correlating zip codes with urban areas, providing first-ever geospatial engagement metrics for university leadership.

4. Selected Project Experience (ICR-Relevant)

Population Raster Algorithm — Core Data Layer for 45+ Country Platform

Scale	45+ African countries; hundreds of active platform users
Stack	Python, R, AWS; grid-based raster processing architecture
Relevance	ICR analytics layer: population data infrastructure, data normalisation for warehouse integration

- Designed and implemented the foundational population raster algorithm migrating legacy hexagonal models to a high-performance grid-based architecture — the data layer underpinning all downstream analytics across Crosscut's platform, directly feeding into the same infrastructure the ICR will leverage for population-denominated campaign coverage analysis.

Geospatial Data Pipelines — 45+ African Country Coverage

Scale	45+ Sub-Saharan African countries
Infrastructure	AWS ECS, Python, R, Docker; 99%+ uptime SLA
Relevance	ICR Phase 5: analytics add-on, FHIR data warehouse synchronisation, structured data accessibility
• Engineers and maintains AWS data pipelines processing geospatial datasets across 45+ African countries with 99%+ uptime, directly relevant to the ICR's requirement for reliable, automated synchronisation of structured campaign data to a data warehouse for external analytical use.	

5. Technical Tools & Open Source

- Contributes to open-source repositories at github.com/crosscutio, including geospatial data pipeline and platform tooling supporting Crosscut's Digital Public Good.
- QGIS plugin development (Python) for geospatial workflow automation.

6. Technical Expertise

Languages

- Python, R, JavaScript, HTML/CSS, SQL, Rust

Infrastructure & Frameworks

- AWS (ECS, S3, and related services), React, Git, Docker, QGIS

7. Education

B.S., Computer Science | Calvin University | May 2024 | Magna Cum Laude (GPA: 3.80)

Appendix 2 – UNICEF Information Security and Platform & Infrastructure Requirements

NOTE: Architecture Summary: All ICR data is stored in the Google Cloud Healthcare API (managed FHIR server). Cinder (web client) and OpenFn (integration middleware) are hosted on GCP VMs. Keycloak provides identity and access management for in-country solutions. For in-country deployments, Ona will follow best practices within the constraints of national data center infrastructure.

Table A – Security Requirements

Requirement	Compl ed?(Ye s/No)	Proposer's Response / Reference
General Security Requirements		
Assessment of, or validation of connectivity controls between UNICEF and any third party / service provider are required before any physical or logical connection can be made. Internally hosted systems that connect and access a Cloud system will use UNICEF standard network deployment topology for the following system components: • For frontend systems • In between systems / containers • Databases End point security controls will be implemented similar if not the same to the ones listed for the top 20 CIS controls. These controls shall be kept functional and updated considering the following domains: • Virus protection • Data exfiltration • Unauthorized access to and/or changing of critical system(s)/application files • Zero-day vulnerabilities		
Vendor agrees to the rights of UNICEF to assess the quality and accurateness of outsourced software development and operational maintenance of the system / application; whether it be through security assurance testing or through external security assessment.	Yes	Ona agrees. The ICR is developed as open-source software (GitHub), and UNICEF is welcome to conduct security assurance testing, code audits, or engage third-party assessors at any time. Ona will provide full cooperation, access to repositories, and documentation to support any such assessment. Ona will coordinate with UNICEF ICTD to validate all connectivity controls prior to establishing any connection. The ICR's data layer is the Google Cloud Healthcare API, a fully managed, HIPAA- and HITRUST-certified service. All connectivity to the Healthcare API is governed by Google's IAM policies and VPC Service Controls. During pilot (cloud-hosted), Ona will complete a formal connectivity review with UNICEF during Phase 2 deployment. For subsequent in-country deployments, connectivity validation will be conducted in coordination with each country's Ministry of Health and UNICEF Country Office, in accordance with national data center and infrastructure requirements.

		The ICR architecture consists of three components, each with a clear security boundary: • Data store (Google Cloud Healthcare API): All patient, campaign, and health data is stored exclusively in the Healthcare API's managed FHIR server. Ona does not operate or manage the underlying database infrastructure — Google handles all storage, replication, and network security for this tier. The Healthcare API is SOC 1/2/3, ISO 27001, HIPAA, and HITRUST certified. • Integration layer (OpenFn): Campaign data from DHIS2, ODK, and other sources is transformed and loaded into the Healthcare API via OpenFn, either using OpenFn's hosted (SaaS) platform or a dedicated GCP VM. OpenFn connects to the Healthcare API via authenticated HTTPS only. • Client application (Cinder): Cinder is a client-side application hosted on a GCP VM that provides the user interface for the ICR. Cinder stores no data; it reads from and writes to the Healthcare API via authenticated API calls. Ona will align the deployment topology with UNICEF's standard network architecture. For in-country deployments, Ona will endeavor to follow these standards while working within the constraints of national data center infrastructure.
Solution / Service shall be protected from unwanted network traffic by network filtering or separating measures that lay outside of the system such as, externally controlled routers and firewalls.	Yes	The ICR is protected by multiple layers of network filtering: • Google Cloud Healthcare API: Protected by Google's global edge network, DDoS mitigation, and API-level access controls. Only authenticated, authorized requests reach the FHIR data store. • Cinder and OpenFn VMs: Protected by GCP VPC firewall rules restricting ingress to only required ports (443/HTTPS) and source ranges. VMs have no public-facing services beyond the application endpoints. • Cloud Armor WAF can be configured in front of the Cinder web application for additional Layer 7 protection if required.
The system shall have proper end-point protection with the following minimum requirements: • Malicious code protection measures • Host firewall configured utilizing, at a minimum, least privileged access controls (services, user, communication access)	Yes	• Malicious code protection: VM instances run hardened OS images with automated security patching. Google's Security Command Center provides threat detection across the GCP project. • Host firewall: GCP VPC firewall rules enforce least-privilege network access for each VM. Only the specific ports and protocols required for each component are permitted. Cinder and OpenFn VMs communicate only with the Healthcare API and Keycloak — no other outbound connections are allowed.
Validation of Security Controls		
Impartial security and vulnerability assessment testing shall be performed to determine the effectiveness of security controls for: • Frontend systems • In between systems / containers • Databases Independent penetration testing agent or teams will perform penetration testing on information systems prior to production release.		

Vendor agrees to the rights of UNICEF to periodically validate the implementation of the security requirements outlined in this document via: • Security Assurance Testing • Vulnerability Testing • Penetration Testing • Audits • On-site checks	Yes	Ona agrees to UNICEF's right to periodically validate all security requirements through any of the listed methods. Ona will cooperate fully with UNICEF-initiated or UNICEF-contracted assessments. The ICR source code is open-source and continuously available for inspection on GitHub.
Compliance & Certifications (OPTIONAL)		
Vendor shall carry ISO2700K certification and/or SOC-2 audits as mandatory or other international standards. The Vendor shall provide information about the certification and/or the SOC-2 report and audit findings or equivalent certifications and/or reports.	Yes	Ona does not currently hold ISO 27001 or SOC-2 certification at the organizational level. However, the ICR's data layer — the Google Cloud Healthcare API — is certified under ISO 27001, SOC 1/2/3, HIPAA, and HITRUST CSF. Since all data is stored and managed within the Healthcare API (not on Ona-managed infrastructure), the most critical security controls are inherited from Google's certified platform. Ona's internal development and operational practices are aligned with ISO 27001 controls. For in country implementations, Ona will work with MoH counterparts to ensure we align with their security and compliance standards as applicable. [GCP Healthcare API compliance: https://cloud.google.com/healthcare-api/docs/concepts/compliance]
Identification, Authentication and Authorization		
These new groups/roles created must follow the principle of least privilege requiring that these groups/roles be given the minimum privileges necessary for their job function.	Yes	The ICR implements role-based access control (RBAC) via Keycloak. All roles are defined following the principle of least privilege. The Google Cloud Healthcare API enforces additional access controls at the FHIR resource level, ensuring each authenticated user or service account can only access the specific data types and operations permitted by their role.
By default, all information systems will rely on existing organization wide security groups and roles that exist on UNICEF's Identity Access Management Services. In cases where applications/services permission cannot be matched to existing groups/roles, a request for new application specific groups/roles creation must be submitted for: • Frontend systems • External partner / other systems / accounts • In between systems / containers (including Staging area) • For privileged accounts accessing the backend systems such as the databases	Yes	Keycloak supports federation and identity brokering. Ona will configure Keycloak to integrate with UNICEF's Identity Access Management system (e.g., via SAML 2.0 or OpenID Connect) so that existing UNICEF security groups can be mapped to ICR application roles. Where ICR-specific roles are needed (e.g., Campaign Manager, Data Analyst, System Administrator), Ona will submit role definitions to UNICEF ICTD for approval before creation. For in-country deployments, roles will also accommodate MoH and partner staff as defined in coordination with UNICEF Country Offices.

All UNICEF staff accounts shall be created on an automated fashion, based on the connection between the IAM System.	Yes	Since this is a facing solution this is not applicable.
The service provider shall follow the principle of least privilege, guaranteeing that users, group, role, and device identifiers will be unique, assigned to each entity (user or process). Each application user role shall have a correspondent database connection according to its privileges.	Yes	Each user and service account in the ICR will be uniquely identified. Application-level RBAC (Keycloak) maps to FHIR resource-level access controls enforced by the Google Cloud Healthcare API. Service accounts used by OpenFn and Cinder have scoped IAM permissions limited to the specific Healthcare API operations they require. No shared or generic accounts are used.
The service provider shall centrally manage the user account using federated identities and whenever possible integrate their solution with the UNICEF Identity Management System. In case authentication is password based; the password shall forcefully adhere to the common best practice quality requirements and will be forcefully renewed frequently.	Yes	Since this is a facing solution this is not applicable.
Multi-factor authentication will be used for: • Privileged accounts and • User access outside of UNICEF trusted network	Yes	Keycloak supports multi-factor authentication (MFA) including TOTP (e.g., Google Authenticator) and WebAuthn. MFA will be enforced for: • All administrative and privileged accounts (system admins, GCP project owners).
All the user and system accounts shall be disabled after a defined period of inactivity, in accordance with organizational standards. All default accounts and/or passwords shall be removed or changed. Approvals will be required for creation, deletion, or modification of any account.	Yes	Keycloak supports session timeout and account inactivity policies. Accounts inactive beyond a configurable threshold will be automatically disabled. All default credentials are changed during initial deployment. Account lifecycle management (creation, modification, deletion) follows an approval workflow documented in the ICR operations manual.
Account lockout features will be used for invalid authentication attempts.	Yes	Keycloak's brute-force detection can be enabled, providing automatic temporary account lockout after a configurable number of failed login attempts. Lockout duration increases progressively. Administrators are notified of repeated failed authentication events.
Application code shall never contain any credentials.	Yes	No credentials, API keys, or secrets are stored in application source code. All secrets are managed via Google Cloud Secret Manager and injected at runtime. GCP service account keys for Healthcare API access use workload identity or short-lived tokens rather than stored key files. The CI/CD pipeline includes automated secret scanning (e.g., git-secrets) to prevent accidental credential commits.
Availability and Deletion		
The system will have a HIGH Level of Availability. Availability requirements and recovery mechanisms shall comply with UNICEF BCM requirements covering critical business processes for: • Frontend systems • In between systems / containers • Databases	Yes	The ICR's availability is anchored by the Google Cloud Healthcare API, which provides a 99.95% uptime SLA with built-in replication, automatic failover, and managed backups: • Data store (Healthcare API): Fully managed by Google with multi-zone redundancy. No single point of failure. Google handles all backup, replication, and disaster recovery for the FHIR data store. • Cinder VM: Can be redeployed rapidly from infrastructure-as-code. Since Cinder stores no data, recovery is straightforward — provision a new VM and redeploy.

		• OpenFn: If using OpenFn's hosted SaaS, availability is governed by OpenFn's SLA. If self-hosted on a GCP VM, the same rapid redeployment applies. RPO and RTO will be aligned with UNICEF BCM requirements during the deployment planning phase. For in-country deployments tied to national data centers, Ona will work with UNICEF Country Offices to achieve the best feasible availability within local infrastructure constraints.
For the deletion of data, a process is required for deletion of any information system data.	Yes	The Google Cloud Healthcare API supports both logical deletion (soft delete with audit trail) and permanent deletion of FHIR resources. Deletion requests are handled through authorized API calls with appropriate access controls. Since Cinder stores no data and OpenFn processes data in transit only, deletion is managed centrally at the Healthcare API layer. All deletion actions are logged.
Any deletion of confidential / personal data must be done so that it cannot be reconstructed.	Yes	When permanent deletion of personal data is required, Ona will perform hard deletion via the Healthcare API's FHIR delete operations, which purge data from the underlying storage. Google's storage systems perform cryptographic erasure, ensuring deleted data is irrecoverable. Deletion procedures will align with UNICEF's data retention and disposal policies. [Reference: https://cloud.google.com/security/deletion]
Cryptography		
Strong cryptographic algorithms shall be used for encrypting data at rest with a minimum standard of AES 256, or in accordance with existing industry best practice or published UNICEF standard, whichever is more restrictive. Data on the following systems must follow the above minimum standards while in transit and at rest for: • Frontend systems • In between systems / containers • Databases The organization will escrow encryption keys to maintain the availability of information in the event of loss of encryption keys. Where applicable, key management guidelines from a trusted Cloud service provider will be followed.		
The service provider shall use best practice or industry standard secure data exchange protocols and keep them up to date. Outdated and/or compromised protocols shall never be used.	Yes	All data exchange uses current, industry-standard protocols: • HTTPS with TLS 1.2+ for all API communications (Healthcare API FHIR endpoints, Keycloak OIDC, DHIS2 integration, OpenFn data flows). • The Healthcare API enforces TLS and rejects connections using deprecated protocols. • Deprecated protocols (SSLv3, TLS 1.0/1.1, RC4, 3DES) are not supported. Protocol configurations are reviewed and updated as part of regular security maintenance.
All passwords shall be encrypted with best current practices or strong industry standards cryptographic algorithms and secure keys.	Yes	Passwords stored in Keycloak are hashed using PBKDF2-SHA256 with a high iteration count and unique per-user salt. Passwords are never stored in plaintext or reversible encryption. Service account credentials for the Healthcare API use Google-managed keys or short-lived OAuth 2.0 tokens rather than static passwords.
Key files must be protected from unauthorized modification using an application that enforces automatic reconciliation from an authoritative source.	Yes	All infrastructure configuration is managed as code (e.g., Terraform) stored in version-controlled Git repositories. Changes require peer-reviewed pull requests and are deployed via CI/CD pipeline. Any drift from the declared configuration is detected and

		reconciled. Secrets are sourced from Cloud Secret Manager; Healthcare API encryption keys are managed by Google Cloud KMS.
Encryption keys shall be securely stored outside of the systems on which they are used.	Yes	Healthcare API encryption keys are managed by Google Cloud KMS, which stores key material in FIPS 140-2 Level 3 validated hardware security modules (HSMs), physically and logically separated from the systems that use them. Application-level secrets are stored in Cloud Secret Manager, separate from the VMs that consume them.
Secure Development (if applicable)		
The system shall be developed following the 'data protection by design and by default' principle.	Yes	The ICR is a reference FHIR implementation and will use Google's Healthcare API as the data store — it provides built-in HIPAA compliance, audit logging, and fine-grained access controls by design. Data minimization, purpose limitation, and role-based access controls are embedded in the platform architecture.
A documented policy and/or process that indicates the use of a known secure development methodology throughout the entire life cycle of the software (SDLC).	Yes	Ona follows a secure SDLC that includes: threat modeling during design, secure coding guidelines, mandatory code review for all changes, static analysis (SAST) and dependency scanning in CI/CD, dynamic application security testing (DAST) before production releases, and security-focused QA testing. The project will maintain public contribution guidelines and code review standards on GitHub.
The security assurance code must be done using the international and best practices in security, such as OWASP, PCI, others.	Yes	Security testing follows OWASP Testing Guide methodology. The CI/CD pipeline includes automated SAST and DAST scans. UNICEF's InfoSec team will be notified before and after QA security assurance testing is performed on each release.
The system shall be engineered following the 'security by design' principles.	Yes	Security by design is a core principle of the ICR architecture. Defense in depth is implemented across layers: authentication (Keycloak + MFA), authorization (Healthcare API FHIR access controls + Keycloak RBAC), encryption (TLS + AES-256 via Google), network segmentation (VPC firewall rules), and audit logging (Healthcare API audit logs + Cloud Audit Logs).
Development and tests of the system will be done with fictitious or pseudonymized information.	Yes	All development, staging, and test environments use synthetic or pseudonymized data. No real patient or beneficiary data is used outside of production environments. Test data generation follows FHIR Synthea-based approaches to create realistic but entirely fictitious campaign data.
Access to program source code and associated items shall be strictly controlled; to prevent the introduction of unauthorized functionality.	Yes	Source code is hosted on GitHub with branch protection rules requiring: pull request approval from at least one authorized reviewer before merge, passing CI checks (tests, linting, security scans), and signed commits where possible. Direct pushes to main/production branches are prohibited. Access to repositories is governed by team-based permissions with regular access reviews.
The system shall display generic error messages that do not disclose detailed information such as process logs, account or system information.	Yes	Cinder and the ICR API layer are configured to return generic, user-friendly error messages in production. Detailed error information (stack traces, system paths) is logged server-side only via Cloud Logging and never exposed to end users. The Healthcare API itself returns standardized FHIR OperationOutcome responses without internal system details.

Executable code will not be implemented on an operational system until evidence of conforming to the testing criteria is acquired and the associated program source libraries have been updated.	Yes	All code deployments to production follow a gated CI/CD pipeline that requires: passing unit and integration tests, passing security scans (SAST/DAST), QA sign-off or User Acceptance Testing (UAT) approval, and successful deployment to a staging environment. No code reaches production without passing all gates.
Security Operations		
Based on UNICEF's security program requirements, all systems shall follow ICTD and the SOC procedures and provisions to maintain a secured network of interconnected devices internally and externally for: • Frontend systems • In between systems / containers • Databases	Yes	Ona will align the ICR's security operations with UNICEF ICTD and SOC procedures. The Healthcare API's infrastructure security is managed by Google's dedicated security operations team with 24/7 monitoring. For the application layer (Cinder, OpenFn, Keycloak), Ona's operations team will follow documented incident response and escalation procedures coordinated with UNICEF's SOC.
UNICEF / third party hosted solutions shall incorporate appropriate controls to meet this requirement.	Yes	The ICR incorporates controls at every layer: network security (VPC firewalls), application security (Keycloak authentication and authorization), data security (Healthcare API encryption and access controls), and monitoring (Cloud Audit Logs, Security Command Center). Controls are documented and available for UNICEF review.
The system shall be hardened, which means that: • Only the services and network ports necessary for efficient operation are up and running • All application code is patched and kept up to date • Limiting the accounts and removing, changing, or disabling default accounts and passwords	Yes	• Minimal services: VMs run only the required application processes. Only port 443/HTTPS is exposed externally. The Healthcare API is a fully managed service with no user-configurable attack surface. • Patching: VMs use GCP's OS patch management for automated security updates. Application dependencies are updated via dependency scanning on a regular cycle. Critical patches are applied within 48 hours. • Accounts: All default accounts and passwords are removed or changed during initial provisioning. GCP service accounts follow least-privilege IAM scoping.
Servers and applications shall be configured to run with the minimum system authorizations necessary.	Yes	VM instances run application processes under dedicated non-root service users. GCP IAM service accounts are scoped to the minimum required permissions (e.g., Cinder's service account can read/write FHIR resources but cannot modify IAM policies or access other GCP services). The Healthcare API enforces resource-level access controls.
The production environment shall be separated from the test and development environments; preferably on logically and physically different systems.	Yes	Production, staging, and development environments are hosted in separate GCP projects with distinct VPCs, service accounts, and access controls. Each environment has its own Healthcare API dataset, Keycloak realm, and VM instances. There is no network connectivity between production and non-production environments.
Development and test environment shall have the same patch level as the production environment.	Yes	All environments are deployed using the same infrastructure-as-code templates and application images, ensuring identical patch levels. Staging is updated before production, serving as a validation gate.
The production environment shall not have any development tools.	Yes	Production VMs do not include compilers, debuggers, package managers, or development tools. Only the application runtime and its dependencies are present.

Configuration/Application source code / customized work shall be protected from unauthorized access / modification and reside in non-production environment with proper back-up / resiliency policy.	Yes	Source code is stored in GitHub with branch protection and access controls. Infrastructure configuration is managed via version-controlled code with full audit history. Production data is backed up via Healthcare API's managed backup capabilities. All configuration changes are tracked.
The system shall have malicious code protection measures. Logs generated by malicious code protection measures shall be monitored.	Yes	GCP's Security Command Center provides threat detection and vulnerability scanning across the project. VM instances use automated OS security updates. The Healthcare API's managed infrastructure includes Google's proprietary malware and intrusion detection. All security events and alerts are logged in Cloud Logging and monitored by Ona's operations team.
Vulnerability Management		
Assessment of, or validation of connectivity controls between UNICEF and any third party / service provider are required before any physical or logical connection can be made for: • Frontend systems • In between systems / containers Patch management of vulnerable system shall be consistent with UNICEF ICTD patch management process.		
The service provider is required to report on the results of the security scans and the remediations taken. Any vulnerability identified that affects the scope shall be reported to the UNICEF focal point.	Yes	Ona will provide regular vulnerability scan reports to UNICEF's designated focal point where applicable. Critical and high-severity vulnerabilities will be reported immediately upon discovery along with a remediation plan and timeline. A summary of all scan results and remediations will be included in quarterly status reports.
The service provider is required to run security tests. Tests will run prior to the launch of the system and periodically afterwards; with a minimum frequency of once a year.	Yes	Security testing will be conducted: • Prior to initial production launch (penetration test + vulnerability assessment). • Before each major release. • At minimum annually via an independent third-party penetration test. • Continuously via automated SAST/DAST in the CI/CD pipeline. The Healthcare API's infrastructure is continuously tested by Google as part of their security program.
The service provider is required to report on the results of the security scans and the remediations taken. These reports will be sent to UNICEF's Chief of IT Security or the relevant focal point(s).	Yes	All security scan and penetration test reports, including findings, severity ratings, and remediation actions taken, will be sent to UNICEF's Chief of IT Security or designated focal point(s) within 5 business days of report completion.
Critical security patches shall be applied following established testing / change management processes.	Yes	Critical patches are fast-tracked through an expedited change management process: patch is tested in staging, validated, and deployed to production within 48 hours. All critical patch deployments are documented with pre/post validation results and communicated to UNICEF. Healthcare API patches are handled automatically by Google.
Change Management		
Changes to the system and/or application post-baseline will be documented (i.e. ver. / build number, etc.), along with description via a formal UNICEF change management process for: • Frontend systems		

• In between systems / containers • Databases		
Any changes to UNICEF system(s) or software shall be agreed upon between ICT and the business division / office owner of the affected system and third party.	Yes	This will not be UNICEF systems in country systems so not applicable. All changes are tracked via GitHub (version-controlled commits, tagged releases with semantic versioning) and a project management tool. Each release includes a changelog documenting changes, version number, and deployment date. Ona will follow UNICEF's formal change management process for production deployments.
Changes to system and/or application post baseline will be documented (version / build number), along with description via a formal change management process. The service provider shall report: type, version, reason, post test results after implementation. Patches that fail testing will also be recorded and documented.	Yes	Each deployment includes documentation of: change type (feature, bugfix, patch, configuration), version/build number, reason for change, test results (pre and post deployment), and rollback status if applicable. Failed patches are logged with root cause analysis and re-test plans. Change logs are available to UNICEF upon request.
The updating of the operational software, applications and program libraries will only be performed by trained and qualified administrators upon appropriate management authorization.	Yes	Production deployments are executed only by authorized Ona DevOps engineers via the CI/CD pipeline. Manual access to production VMs and GCP resources requires MFA and is restricted to a small group of trained administrators. All production actions are logged via Cloud Audit Logs.
Log Management and Monitoring		
Logs/events will be generated in a format that can be easily parsed and used as an input for logging process management for: • Frontend systems • In between systems / containers • Databases		
Event filtering will be enabled on logs for analysis to fine tune the results and consolidate logs into single entry containing count of number of occurrences of events to save space. Analysis will be performed on communication patterns to generate profiles for information systems for detecting unusual activities in the network.		
The system shall generate and process auditing tracks covering all actions taken on personal data, including data access only.	Yes	The Google Cloud Healthcare API maintains a complete audit trail of all FHIR resource operations (create, read, update, delete) with user attribution, timestamp, and resource identifiers via Cloud Audit Logs (Data Access logs). This covers all actions on personal data including read-only access. Keycloak logs all authentication and authorization events. Together these provide a comprehensive audit trail.

Authentication validation activities and all changes in authorization shall be logged and securely stored, with limited access.	Yes	Keycloak logs all authentication events (successful logins, failed attempts, token issuance) and authorization changes (role assignments, permission modifications). Logs are stored in Cloud Logging with access restricted to authorized administrators only. Log data cannot be modified or deleted by application users.
Access to content, key information and/or any modifications to operational program libraries shall be logged and restricted.	Yes	Access to production configuration, encryption keys (Cloud KMS), and GCP resources is logged via Cloud Audit Logs (Admin Activity and Data Access logs). Access is restricted to authorized personnel via IAM policies with MFA enforcement.
Logs and events will be generated in a format that can be easily parsed and used as an input for logging process management.	Yes	All logs are emitted in structured JSON format, compatible with standard log management and SIEM tools. Logs can be exported via Cloud Logging sinks to BigQuery, Pub/Sub, or Cloud Storage for integration with UNICEF's logging infrastructure if required.
Integrity log checking shall be performed to ensure consistency.	Yes	Cloud Logging provides tamper-evident log storage. Cloud Audit Logs are immutable and managed by Google. Critical audit logs can also be exported to locked Cloud Storage buckets with retention policies to prevent tampering.
The system, application, as well as underlying services and/or networks, shall be monitored and activities logged.	Yes	Comprehensive monitoring is implemented via Google Cloud Monitoring covering: • Infrastructure: VM health, CPU, memory, disk, network metrics. • Application: Cinder and Keycloak health, API response times, error rates. • Healthcare API: FHIR operation latency, error rates, quota usage. • Security: Failed auth attempts, firewall denials, anomalous traffic. Alerts are configured for critical thresholds with notification to the Ona operations team.
Security Incident Management		
Reporting of security incidents shall be in coordination with the SOC whether internally or externally. A security breach shall be viewed as: • A failure in security controls which leads to the accidental, unlawful, or unauthorized access, destruction, loss or alteration of data/information processed/stored on the system • A failure in security controls which leads to the accidental, unlawful, or unauthorized access to ICT resources		
Security breaches shall immediately be communicated to UNICEF's Point of Contact.	Yes	Ona commits to immediately notifying UNICEF's designated Point of Contact upon confirmation of any security breach. Initial notification will include: nature of the breach, systems/data affected, containment actions taken, and estimated timeline for full assessment. Follow-up reports will be provided as the investigation progresses.
A security incident notification and escalation procedure shall be formally documented and contractually enforced between the service provider and UNICEF's Security Operations Centre.	Yes	Ona will work with UNICEF to develop and formally document a security incident notification and escalation procedure prior to production deployment. This procedure will define severity levels, notification timelines, escalation paths, communication channels, and responsibilities.

Table B – Platform and Infrastructure Requirements

Requirement	Complied?(Yes/No)	Proposer's Response / Reference
The Vendor shall integrate their solution with the established authentication and identity framework; implementation / adoption of Single Sign-On when available.	Yes	The ICR uses Keycloak as its identity provider, which natively supports SSO via SAML 2.0 and OpenID Connect.
Documented firewall considerations and required configurations.	Yes	Firewall configurations are managed as code and include: • GCP VPC firewall rules (ingress/egress restrictions by port, protocol, and source range for Cinder and OpenFn VMs). • Cloud Armor WAF policies can be configured for additional Layer 7 protection. • Healthcare API access is controlled via IAM and OAuth 2.0 scopes, not network firewalls — only authenticated, authorized requests are processed. Documentation of all firewall rules and their rationale will be provided to UNICEF as part of the deployment package.
Vendor to provide a clear, publicly stated stance on privacy protection and data use, re: Privacy Policy.	Yes	Ona maintains a public privacy policy (available at ona.io). Ona commits to processing data under this engagement solely for the purposes defined in the contract with UNICEF. No data processed through the ICR will be used for any other purpose, including Ona's own analytics, marketing, or product development.
Documented high-availability and disaster recovery capabilities and procedures.	Yes	The ICR's HA/DR capabilities are anchored by the Google Cloud Healthcare API: • Healthcare API: Fully managed with multi-zone redundancy, automatic failover, and 99.95% uptime SLA. Google handles all backup and disaster recovery for the FHIR data store. • Cinder / OpenFn VMs: Stateless (no data stored); can be redeployed from infrastructure-as-code within minutes. VM snapshots provide additional recovery options. • Documented DR runbook covering component failure scenarios will be provided. For in-country deployments, HA/DR will be adapted to available local infrastructure.
Adherence to GDPR or DoD 5220.22-M or NIST SP 800-88 for data sanitization on retirement of storage devices.	Yes	GCP handles physical storage device retirement and sanitization in compliance with NIST SP 800-88 guidelines. Google's data destruction process uses cryptographic erasure and physical destruction of storage media. Since all ICR data resides in the Healthcare API (no local data on VMs), data sanitization is fully covered by Google's processes. [Reference: https://cloud.google.com/security/deletion]
Publish/provide locations of data centers storing data.	Yes	For cloud-hosted piloting, the ICR will be hosted in a GCP region selected in consultation with UNICEF, considering data sovereignty requirements of the pilot countries. Initial candidate regions include europe-west1 (Belgium) or a region closer to the pilot countries. Specific data center locations will be documented and confirmed with UNICEF prior to deployment. For in-country deployments, data center locations will be determined by national infrastructure and data sovereignty requirements, in coordination with UNICEF Country Offices and Ministries of Health.

Documentation of data retention procedures and backup facilities.	Yes	Data retention and backup procedures: • Healthcare API FHIR data store: Automated backups managed by Google with configurable retention. • Application logs: Retained in Cloud Logging per UNICEF's requirements (configurable). • Audit logs: Retained for the duration of the engagement plus a post-contract retention period agreed with UNICEF. • Backup storage is encrypted (AES-256) by Google. Full data retention SOP will be provided.
Business Continuity Plan including backup and restore procedures.	Yes	Ona will develop and maintain a Business Continuity Plan (BCP) for the ICR covering: backup and restore procedures (tested periodically), failover processes, communication plans during outages, and recovery priorities aligned with UNICEF's BCM requirements. The BCP will be reviewed and updated annually or after any significant incident.
Describe in detail all the annual support and maintenance schemes that they provide.	Yes	Ona's annual support and maintenance includes: • Technical support via email and ticketing system during business hours (with critical issue escalation 24/7). • Regular software updates (security patches, bug fixes, feature enhancements). • Periodic system health reviews and performance optimization. Detailed support SOP will be included in the deployment documentation.
Clearly define the procedure to handle escalation issues, bugs, and service packs.	Yes	Escalation procedure: 1. Issues reported via ticketing system with severity classification. 2. Triage by Ona support team within SLA response time. 3. Escalation path: Support Engineer → Senior Engineer → Technical Lead → CTO. 4. UNICEF focal point notified of critical/high severity issues within 4 hours. 5. Hotfixes follow expedited change management with staging validation before production deployment. 6. Post-incident reports provided for all critical issues.

Table C – Privacy and Data Protection Requirements

Requirement	Compl ed?(Ye s/No)	Proposer's Response / Reference
Privacy Governance and Management		
Proposer has implemented a comprehensive privacy program to comply with applicable data protection laws, regulations and international best practices of personal data protection, including a personal data breach procedure. Kindly identify which framework was used: • GDPR • Generally Accepted Privacy Principles (GAPP) • HIPAA • UN Privacy Principles • Other (please clearly identify)	Yes	Ona's privacy program is informed by: • GDPR: As a company working with international organizations and processing data of individuals globally, Ona's data handling practices align with GDPR principles (lawful basis, data minimization, purpose limitation, data subject rights). • UN Privacy Principles: Given Ona's extensive work with UN agencies (UNICEF, WFP), our practices are aligned with the UN Personal Data Protection and Privacy Principles. Ona follows a data breach response procedure that includes detection, assessment, notification (to data controllers and relevant authorities), containment, and remediation.
Proposer oversees and enforces compliance with the privacy programme. Kindly describe how such activities are carried out and specify the frequency and audit activities performed.	Yes	Privacy compliance is overseen by Ona's management team with support from legal counsel. Key activities include: • Privacy impact assessments for new projects involving personal data. • Annual review of data handling practices and privacy policies. • Staff training on data protection during onboarding and annually thereafter. • Code review processes that include privacy-relevant checks (data minimization, access controls, logging of personal data access).
Proposer agrees not to use any data (even if anonymized) processed for any other purpose than for the purposes of the Proposal (e.g., AI/machine learning, deriving other data, etc.)	Yes	Ona agrees. All data processed through the ICR will be used exclusively for the purposes defined in the UNICEF contract. Ona will not use any ICR data — including anonymized or aggregated data — for its own research, product development, AI/ML training, marketing, or any purpose outside the scope of this engagement.
The solution complies with best industry standards of "privacy by design and default".	Yes	The ICR is designed with privacy by design and default: • Data minimization: Only data necessary for campaign operations is collected. • Centralized data store: All personal data resides in the Google Cloud Healthcare API, which provides built-in HIPAA compliance and fine-grained access controls. Cinder (client) and OpenFn (integration) store no personal data. • Access controls: RBAC via Keycloak ensures users only access data relevant to their role. • Pseudonymization: The FHIR data model supports patient identifiers separate from personal details. • Encryption: All data encrypted in transit (TLS) and at rest (AES-256) by Google. • Audit trail: All access to personal data is logged via Healthcare API audit logs.

Describe the implementation of an easily visible and understandable appropriate “opt-out” mechanism when collecting consent.	Yes	The ICR will implement consent management aligned with the specific requirements of each pilot country’s data protection regulations. The FHIR data model includes a Consent resource that can record and enforce consent status per individual. Where applicable, the data collection interface (e.g., OpenSRP mobile app, ODK forms) will include clear, user-friendly consent screens in local languages with an opt-out option. Consent status is stored as a FHIR Consent resource in the Healthcare API and enforced at the data access layer.
--	-----	--